# Active-Rail Spacetime Metric Engineering System
## Technical Disclosure and Source-Placement Architecture

**Engineered Operating Embodiment: Service Factor $V = 5$**
**Service-Parameter Diagnostics: $V = 2.5$ and $V = 10$ Outside Current Engineered Scope**

**Document purpose.** This document discloses the representative active-rail metric-service architecture, its source-role decomposition, the archived beta075 endpoint and exchange models at $V = 5$, and the requirements for a coupled physical source construction. It distinguishes prescribed-metric service results, fitted endpoint tensors and exchange currents, fixed-background effective-operator estimates, and independently supplied material and quantum stresses. The joint construction coordinates standing support, angular response, transport, storage, finite transitions, and quantum backreaction through common geometry and explicit exchange laws.

## 1. ABSTRACT

An active-rail spacetime metric engineering system is disclosed as a staged transport service built from prepared infrastructure, protected packet motion, timed handoff, source placement, carrier auditing, and chronology governance. Standing radial support, angular capacity, endpoint recovery, release timing, and reset behavior have distinct responsibilities in the surrounding plant. Their coordinated action preserves the protected corridor in the representative prescribed-metric service tests.

The representative $V = 5$ package supplies a prescribed operating regime, measured service and carrier margins, a source-role ledger, a fitted anisotropic endpoint tensor, and a complementary exchange-current model. Its effective transport operators have fixed-background characteristic and energy estimates. Physical realization requires a pointwise tensor for every supporting sector, corrected causal-frame checks, and simultaneous closure of material, quantum, and Einstein equations. The source tests distinguish bulk radial tension from negative null stress and connect both to angular response, transition forces, and stored initial state. This decomposition defines the coupled source-selection problem and its bounded acceptance sequence.

## 2. TECHNICAL FIELD

The disclosure relates to prescribed spacetime metric engineering, numerical relativity scaffolding, source-placement architectures, and controlled metric-service systems. More particularly, the disclosure relates to a rail-like spacetime metric in which prepared infrastructure bears broad geometric support while live service demand is routed through localized handoff windows and packet-adjacent control regions.

## 3. ENGINEERING OBJECTIVE

A practical spacetime metric engineering architecture benefits from explicit source placement, explicit service timing, and explicit causal-carrier classification. The disclosed system assigns standing radial support, throat/angular support, clock-lapse safety, carrying-flow load, entry service gating, catch/rematch correction, packet release, endpoint receiver action, support-shell timing, and reset operations to distinct service components. The engineering objective is to provide a reduced metric and source-placement process that:

1. separates prepared substrate support from live transport increments;
2. keeps the protected packet corridor causally safe throughout live service;

3. localizes small service-induced momentum demand into catch/rematch and support-shell handoff regions;
4. expresses the reduced metric in ADM variables for constraint-ledger diagnosis;
5. subtracts the standing substrate from live service fields to isolate transport demand;
6. characterizes 4D demanded-source responses for load-bearing component assignment;
7. supports prescribed reduced geometry, source-family target realization, endpoint source-medium closure diagnostics, support-stress exchange closure, fixed-background source-family equation validation, fixed-background energy-estimate certification, causal-carrier auditing, and coarse 3+1 constraint accounting within one design grammar.

## 4. OVERVIEW OF THE ACTIVE-RAIL SYSTEM

The active rail operates as a coordinated metric service. A prepared standing substrate supplies the baseline throat, support shell, angular jacket, rail stretch, and clock-lapse structure. A live packet corridor lies inside the support shell. A carrying-flow field expresses the commanded service load through the prepared support infrastructure. The entry service gate assigns setup support to infrastructure accounting and begins live packet scoring inside the supported entry corridor. A locked timing layer coordinates support-side catch with packet rematch. A release-choreography layer then holds the packet briefly in a matched state, fades beta through a finite smooth profile, keeps lapse/carve support active through the handoff, and relaxes infrastructure support after the packet edge transfers the principal mismatch into the designated handoff windows. Packet-edge and support-shell control components are applied during catch/rematch and release to address localized handoff demand.

The invention-level structure combines the componentized service architecture, source-target grammar, effective endpoint/exchange equations, source-release timing, and carrier classification. The service factor $V$ rates prescribed operating loads and A-to-B service time against an exterior null signal. The representative $V = 5$ configuration preserves packet safety under its recorded smoothing and grid refinements. The $V = 2.5$ ladder preserves live-packet source safety and fitted endpoint tensor identity but misses the V5 medium/support calibration; the $V = 10$ ladder fails live packet source safety. These service-rating diagnostics retain their respective calibration and causal-margin scope.

The principal non-live radial body has a constant-flux string-cloud realization target. A finite endpoint fit supplies a reset cap, a support-edge component, a counted current regulator, and internal angular response. A $24 \times 14$ support basis fits the complementary endpoint exchange current. Throughout this disclosure, *sealed beta075 V5 reference* denotes the archived metric parameters, fitted tensors, and effective-operator calibration. Its support-closure measurements concern exchange-current residuals; a separately supplied reservoir tensor and the endpoint replacement residual enter the complete physical-source construction. The fixed-background characteristic, transport, and energy estimates remain properties of their stated effective operators. Physical rest-frame admissibility uses the corrected causal classifier and requires reevaluation of the corresponding archived margins.

The source responsibilities overlap through shared geometry and material state. Bulk radial tension, signed radial and angular null stress, current transport, and stored support response retain separate budgets, while their exchange determines interface forces and timing. Section 15 specifies their common accounting and ordered joint construction. The carrier classification continues to evaluate reachability, radial escape, scheduled probes, dense branch-band bundle transport, wake-tail restoration, exterior-null service time, and rail-time chronology governance on the stated prescribed backgrounds.

## 5. REPRESENTATIVE DESIGN ARCHITECTURE

The representative design has five coupled layers. The metric-service layer is the prepared ADM program $(\alpha, \beta, \gamma_{ll}, \gamma_{\Omega\Omega})$ with $V = 5$ as the engineered operating rating. The packet-handoff layer is the matched-hold release choreography with a smooth beta fade, blended trailing-edge beta-rematch sleeve, finite beta-collar widening for branch-band bundle preservation, two-feature packet-local radial support, compact entry/catch/edge handoff support, and entry service gate near $\sigma = -1.40$. The source-family layer is

$$\mathcal{S}_{\text{target}} = S_0 + J + R + C_{\text{live}} + G + D_H, \tag{1}$$

where $S_0$ is the constant-flux radial string-cloud backbone, $J$ is the coupled endpoint/junction layer, $R$ is small core-body residual leakage, $C_{\text{live}}$ is live handoff trim, $G$ is angular-capacity support, and $D_H$ is non-live current relaxation.

The endpoint receiver/source layer uses beta-release memory to drive a localized negative-$l$ angular receiver in the non-live support-edge band. The archived beta075 *p003_mid* reference preserves the recorded live packet gate, improves selected-null transfer, and retains positive affine-reparameterized benchmark margin on its source-sector windows. Its finite endpoint fit combines a bounded reset cap, a finite support-edge component, the current regulator, and internal angular response. Covariant tensor identity verifies this fitted medium. The difference from geometric $J$ remains an explicit realization target.

The $24 \times 14$ support basis supplies the fitted complementary exchange current and its baseline/dense localization, live-exclusion, angular-exchange, and support-tail checks. The cap-0.95 phase-local source law, scoped entry/catch normalization, and service-coordinate timing define the associated effective transport problem. Its observed-amplitude full-domain evolution has an inactive limiter and an aligned-envelope rapidity bound. The director/support-reservoir operator supplies storage, damping, and power/radial-force response variables; a physical reservoir construction additionally supplies its tensor and energy cost. The energy-estimate layer equips the fixed-background operator with a positive energy norm, symmetric principal matrix, in-cone flux accounting, and explicit lower-order work constants. The beta100 release-timing stress condition retains the recorded service-specific receiver retiming and transfer diagnostics.

The reduced support-sector transport layer treats the accepted endpoint/support closure as a fixed-background local constitutive system. The associated action-level matter model is a later derivation layer. Its principal-symbol diagnostic uses primitive perturbations

$$U = (h, \psi, \chi_\Omega, \pi_\Omega, \Phi_{\text{support}}, \Pi_{\text{support}}, n_l)$$

to test heat/enthalpy response, angular response, support stroke/stress response, and director advection. On the promoted $24 \times 14$ endpoint/support reference, the baseline and dense local principal-symbol checks have real finite in-cone characteristics and complete eigenbases with zero hard gate-crossing rows. The dense run has 56 boundary-near rows, minimum cone margin $7.881 \times 10^{-5}$, and p01 cone margin 0.017633. Heat-current sensitivity identifies the reset-decompression/support-edge rows as the local boundary: raw heat-ratio worsening of order $10^{-4}$ reaches the first hard local gate crossing, while bounded transport rapidity preserves the hard gates at the same observed stress. Row-budget, $1 + 1$ advection, full-package source-coupling, scoped support-edge source-profile normalization, and full-domain fixed-background evolution keep the observed-amplitude source-driven rapidity evolution below budget with the limiter guard inactive. The deliberate large-amplitude full-domain stress case also stays within budget in the fixed-background evolution rung. At action-level fixed-background scope, service-coordinate source release supplies the timing grammar: all tested aligned pulse widths pass, and the convex-kernel aligned-envelope certificate bounds any common nonnegative temporal kernel over the service-ordered bins with maximum budget fraction 0.742835. Source-family equation validation recomputes the local principal symbol from the regulated director/support-reservoir equations and preserves the inherited hard gate class on baseline and dense V5 surfaces, with the dense V5 surface retaining the tightest cone and local support-closure margins. The $V = 2.5$ and $V = 10$ ladders are therefore interpreted separately as service-rating diagnostics rather than as current cross-surface proof surfaces. The fixed-background energy certificate organizes the validated source-family equations as a symmetric energy-estimate system with positive identity symmetrizer, symmetric principal block, in-cone flux bound, and explicit lower-order work constants. Bounded service-local timing jitter, energy-constant auditing, and first-order coupled consistency are the current theorem-style extensions for this fixed-background source family.

The causal-carrier layer evaluates the shell/throat carrying-flow feature through a dedicated prescribed-metric carrier audit. The local GZ-like branch signature is measured in the live and shell/throat overlap bands. Service-stage and carry-stage entry-to-live-packet reachability screens record zero hits in the sampled ledgers. Refined beta075 live packet, main carrier, support plant, packet geometry, and live branch-band seeds reach exterior radial boundaries on both radial null branches, including minus-branch traces through the branch-sensitive region. Dense finite-bundle branch-band audits are included with the single-ray and scheduled-probe gates: the selected widened trailing-edge beta-rematch collar keeps dense branch-band bundles radially escaping, recovered from both-shrinking intervals, and above the caustic-like collapse thresholds used in the prescribed-metric audit. The service-time advantage ledger compares the reconstructed restored packet arrival event with an exterior null reference between the same endpoints and rates the representative schedule as operationally favorable. The chronology-governance layer treats each permitted service edge as an advance in a shared rail time with readiness margin. Reset-domain wake-tail monitors remain exterior to live service.

## 6. DEFINITIONS

| Term | Meaning in this disclosure |
|---|---|
| Active rail | A spacetime metric service in which prepared infrastructure supports the transport geometry and a live packet is carried through a protected corridor. |
| Service factor $V$ | A dimensionless carried-flow service setting. It sets the scheduled plateau level of the support-contained carrying-flow field, the packet-frame comparison schedule used for causal-margin accounting, and the direct service-factor proxy used in A-to-B service-time rating. |
| Service-time advantage ledger | An operational exterior-null comparison ledger. It fixes exterior endpoints A and B, defines packet departure and restored arrival events, reconstructs service time through entry, carried shift, catch/release, and restoration, and compares that service time with an ordinary exterior null reference time. |
| Prepared standing plant accounting | A timing convention in which already prepared infrastructure is accounted as standing plant, while request-triggered setup is included in the service-time clock. |
| Rail time $T_{\mathrm{rail}}$ | A service-governor chronology variable assigned to an active rail or connected rail network. Admissible service edges, ordinary causal edges, reset intervals, and return/cross-link edges advance this variable with configured safety margin. |
| Chronology-governed service sequencing | A service-arming rule in which entry, carried shift, catch/release, endpoint readiness, reset, and network routing are admitted according to monotone $T_{\mathrm{rail}}$ advance. It frames closed-timelike-curve exposure as a subsequent safety-engineering layer for service scheduling and network governance. |

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| Engineered operating embodiment | The representative service configuration evaluated and sealed at $V = 5$. |
| Service-rating diagnostic | A fresh same-quality ladder at a service factor outside the engineered $V = 5$ scope, used to map calibration debt or boundary failure. The current diagnostics are $V = 2.5$ and $V = 10$. |
| High-service boundary diagnostic | The $V = 10$ ladder, which fails live packet source safety in the current beta075 package and therefore requires an additional causal-margin engineering mechanism before it can be promoted. |
| Entry service gate | A service-accounting boundary that assigns pre-entry setup support to infrastructure accounting and begins live packet scoring inside the supported entry corridor. |
| Standing substrate | Prepared support geometry including throat/core support, radial support shell, angular jacket, and baseline lapse structure. |
| Live packet | Passenger-facing corridor whose causal and source-exposure properties are monitored during live service. |
| Support shell | Wider infrastructure region enclosing the live packet and carrying broad support demand. In the support-shell control overlays, the active support-shell band is an annular infrastructure band at the support edge. |
| Carrying-flow field | The service-facing name for the radial shift component $\beta$. It carries the scheduled service load through prepared support infrastructure. |
| Clock-lapse field | The service-facing name for the lapse component $\alpha$. It supplies causal margin and can be used as a timed partner to carrying-flow control. |
| Rail-stretch field | The service-facing name for the radial spatial metric component $G = \gamma_{ll}$. It sets radial metric capacity and affects how carrying-flow gradients appear in the source ledger. |
| Throat-capacity field | The service-facing name for the angular metric component $Q = \gamma_{\Omega\Omega}$. It includes throat and angular-jacket capacity. |
| Locked timing | Timing rule in which support-side catch begins with a controlled lead relative to packet rematch while packet-frame safety is constrained. |
| Release choreography | Coordinated matched-hold, beta fade, lapse support, carve support, and packet-edge rematch sequence used to remove carrying-flow support while controlling derivative cost in the live packet. |
| Matched hold | A release-choreography phase in which the packet is kept locally matched for a short interval while the trailing edge and support infrastructure settle. |
| Trailing-edge beta-rematch sleeve | A packet-adjacent beta correction layer, strongest near the inner/trailing packet edge and weak in the packet center, used to soften local $(v_{\mathrm{coord}} + \beta)$ mismatch while preserving a quiet packet center. |
| Finite minus-branch carrier-focus collar | The localized branch-band handoff layer in which the minus radial null family is most sensitive to beta-rematch tuning. In the representative design it is treated as a controlled carrier-rematch collar outside the passenger-facing transport corridor. |

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| Beta-collar rematch lead | A widened trailing-edge beta-rematch collar that preserves packet timelikeness and dense branch-band bundle width in the prescribed-metric carrier audit. The conservative representative lead is the width-6.0, temporal-1.5 trailing-edge rematch setting, with a width-8.0, temporal-2.0 setting retained as a wider optical-relief bracket. |
| Blended beta floor | A smooth union between a weak filled packet-center beta floor and a shaped trailing-edge rematch sleeve. The representative V5 operating embodiment uses this blended floor with a widened sleeve. |
| Two-feature radial support profile | A packet-local $\gamma_{ll}$ support profile using a mild filled core stretch, a narrow annular catch ring, and a weaker annular skirt. It reduces radial-null and angular-pressure burden while preserving packet safety. |
| Catch/rematch absorber | Localized packet-edge source or control component assigned to residual null/momentum handoff demand. |
| Radial support edge | Transition layer controlling radial pressure demand and support-shell coupling. |
| Lapse cushion | Clock-lapse-sector compensator that preserves causal margin consumed by stronger support-edge shaping. |
| Support-shell overlay | A localized metric-side support-shell component applied through a smooth timing and radial window $W_{\mathrm{sh}}(\sigma, l)$. V5 source-ledger studies identify raised-cosine annular support-shell bearing as a clean comparator and reusable control knob; the release-choreography family retains support-shell/lapse infrastructure while focusing selection on packet-edge handoff smoothing. |
| Compact handoff layer | An endpoint-smooth entry/catch/edge support layer that separates entry pressure containment from broad catch/rematch and packet-edge smoothing. |
| Pressure-aware entry profile | A compact entry support function that carries live radial-pressure containment while preserving a separate handoff sleeve for angular/current and radial-null derivative relief. |
| Semilocal null-accumulation diagnostic | A radial-null source diagnostic that averages $T_{kk}$ over finite service/radial windows and compares the result with a selected quantum-inequality-style floor. |
| Affine-reparameterized radial-null diagnostic | A finite-window radial-null screen using an explicitly reparameterized affine window coordinate. It computes radial-null non-affinity, integrates the associated affine weight, and records radial geodesic residual monitors. |
| 4D demanded-source ledger | Finite-difference evaluation of $T^{\mathrm{dem}}_{\mu\nu} = G_{\mu\nu}[g]/(8\pi)$ for the prescribed metric. It diagnoses radial null, radial pressure, angular pressure, current, and packet-comoving density demands. |
| Strong 4D source redistribution | A source-ledger result in which a metric component measurably reduces or reroutes expensive demanded-source channels while preserving packet safety, bounded point peaks, and numerical robustness. |
| Standing substrate subtraction | Diagnostic subtraction of a prepared beta-off or ready-state substrate from live service fields to isolate service-induced ADM demand. |

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| Source-family target | A responsibility map derived from the demanded-source ledger. Physical completion assigns independent tensors, interactions, and full-domain residuals to the named roles. |
| Constant-flux radial string-cloud backbone | The non-live radial-tension target with positive density, $p_l \approx -\rho$, and small current and transverse pressure. Its areal flux is $\Phi = \rho\gamma_{\Omega\Omega}$; the ideal density is $\mu = \Phi/\gamma_{\Omega\Omega}$. |
| Endpoint/junction layer | A coupled non-live source target at support-edge and reset/decompression endpoints. It carries residual radial trim, angular capacity, and current relaxation after subtracting the constant-flux radial backbone. |
| Beta-memory receiver | A support-edge endpoint receiver whose amplitude is tied to accumulated beta-release transfer and expressed as an active endpoint receiver channel. The representative effective receiver uses a negative-$l$ localized angular channel in the endpoint/junction grammar. |
| Frozen endpoint-J source | The beta075 repaired-lead endpoint source package: a bounded reset-decompression cap body paired with a finite support-edge closure component. It is non-live, finite-width, and stable under the baseline and dense source ledgers used for the current disclosure. |
| Sealed beta075 V5 reference | Archived metric parameters, endpoint fit, $24 \times 14$ exchange-current fit, and fixed-background effective-operator calibration at $V = 5$. Its service, transport, and energy-estimate results retain their recorded backgrounds and norms. The physical source assembly includes a separately supplied reservoir tensor and corrected rest-frame checks. |
| Endpoint current regulator | A small non-live regulator in the $\rho+p_l$ radial pair that makes the endpoint radial heat/current block boost-diagonalizable while preserving current, angular, collar, packet, and closure-component exclusion. |
| Regulated anisotropic heat/current medium | The fitted beta075 endpoint tensor with radial heat/current, a counted radial-pair regulator, and internal angular response. Its bounded heat ratio is an effective transport variable; the physical energy frame follows from the corrected timelike eigensystem. |
| Entrained radial director | A preferred radial support direction field paired with the regulated heat/current medium. It supplies the geometric direction for radial heat/current transport, director advection, support exchange, and fixed-background source-family principal-symbol checks. |
| Localized support reservoir | A source target for stored energy, momentum, and strain receiving endpoint power and radial force. The effective model supplies storage/response equations and a fitted exchange current; physical completion supplies a pointwise tensor with that divergence. |
| Regulated director/support-reservoir source family | The endpoint/transport/reservoir construction framework: anisotropic heat/current response, a radial director, stored support state, bounded effective rapidity, and scheduled exchange. Its physical specialization supplies the component tensors, constitutive laws, and reciprocal forces. |
| Formal source-family equation package | The fixed-background endpoint, director, angular-response, and reservoir operator equations, with bounded heat-ratio response, phase-local forcing, scheduled release, positive split-step transport, and a target equation for the reservoir tensor's divergence. |

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| Derivative support-reservoir operator | A fixed-background support operator with storage and stress variables, including $\partial_\tau \Phi_{\text{support}} = \Pi_{\text{support}}$, spatial support derivatives, damping, and lower-order power/radial-force exchange drives. It supplies the derivative form used to validate the support reservoir principal symbol and local exchange-shape margins. |
| Cross-surface source-family robustness | A fixed-background validation of the same regulated director/support-reservoir equations on the available complete engineered source-family surfaces. The current proof surface set is sealed baseline V5 and sealed dense V5, with hard formal-symbol gates, live-exclusion gates, total-conservation gates, and drift gates preserved. Service-rating ladders outside V5 are separate diagnostic maps rather than present proof surfaces. |
| Fixed-background energy-estimate certificate | A symmetric-hyperbolic certificate for the regulated director/support-reservoir source family on the available complete surfaces. It uses an identity block symmetrizer, symmetric principal matrix, positive energy norm, in-cone flux bound, and explicit lower-order work constants for support exchange and scheduled source response. |
| Lower-order energy constant audit | A classification of the energy-estimate work constant into its support-work, local exchange-shape, and support-closure contributions. The current dense V5 value is a stable limiting theorem constant below the hard bound, with explicit headroom and a safety-margin monitor carried into coupled-consistency tests. |
| First-order coupled consistency diagnostic | The reduced constraint-driver and response-proxy extension of the fixed-background source model. A physical Einstein–matter evolution additionally supplies the signed component tensors, their equations, and the metric constraints. |
| Constrained medium field closure | A finite-difference validation in which the frozen endpoint source, endpoint current regulator, and internal angular response close the beta075 endpoint medium with live exclusion and fixed design components. |
| Covariant endpoint/source tensor identity | Reconstruction of the fitted endpoint medium as a covariant tensor. The difference between this fit and geometric $J$ is accounted separately in the full source sum. |
| Support stroke/stress-potential sector | A phase-aware basis and effective state operator used to fit complementary support exchange. A physical specialization evaluates its independent tensor and verifies that tensor's divergence against the fitted current. |
| Total endpoint/support exchange closure | The residual and localization test comparing endpoint divergence with the complementary fitted support current. Full physical conservation additionally evaluates the independent reservoir tensor and all other component exchanges. |
| Compact support-stress bracket | A smaller support-stress basis that preserves aggregate closure and localization and is retained as a compactness bracket when dense local active total-closure selects a larger reference basis. |

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| Reduced endpoint/support principal-symbol diagnostic | A fixed-background local characteristic diagnostic for the endpoint/support constitutive sector. It perturbs $U = (h, \psi, \chi_\Omega, \pi_\Omega, \Phi_{\text{support}}, \Pi_{\text{support}}, n_l)$ and checks real finite in-cone characteristic speeds, eigenbasis completeness, heat/current transport margin, and support-stress response on the accepted audit rows. |
| Bounded transport rapidity | The effective variable $\psi$ defined by $2j_l/\lvert\rho + p_l\rvert = \tanh\psi$ on resolved non-diagonal Type I rows. It parameterizes the heat ratio used by the reduced operator; physical boost rapidity and diagonal degeneracy are treated separately. |
| Rapidity budget diagnostic | A row-local admissibility budget that computes the largest positive $\Delta\psi$ preserving the reduced principal-symbol cone-margin gate. It identifies the tightest reset-decompression/support-edge rows and classifies observed, large-amplitude, and timing-envelope source kicks against the same invariant budget. |
| Rapidity advection non-concentration gate | A reduced $1 + 1$ fixed-background radial advection test for rapidity impulses and source-driven rapidity profiles on service-radial slices. It verifies that observed rapidity sources stay inside row-local budgets under the tested advection law. |
| Support-edge source-profile normalization | A scoped pre-evolution normalization of the support-edge endpoint/junction entry/catch source profile against the observed-amplitude rapidity budget. It keeps the V5 observed full-package source-coupling sweep inside budget using the existing geometry, packet, current, and closure components. |
| Cap-0.95 support-source law | A phase-local normalized source-coupling law for the support-edge endpoint/junction source slices. It is applied within the sealed beta075 endpoint/support package and preserves live exclusion, bounded rapidity budgets, and inactive observed-amplitude limiter behavior. |
| Full-domain fixed-background evolution | A split-step reduced evolution over the active non-live support-source domain using the sealed endpoint/support package, bounded rapidity variable, cap-0.95 source law, and row-local budgets. Observed outward/inward/backward scenarios stay below budget with zero live or packet-live evolved rows. |
| Service-coordinate source-release law | A timing law that releases support-source rows according to service coordinate and service direction. It represents rail operation as a scheduled source process with shared service ordering. |
| Aligned-envelope rapidity certificate | An action-level fixed-background bound showing that the one-step service-aligned basis response bounds any common nonnegative temporal kernel over the service-ordered source bins by linearity and positivity of the split-step transport operator. |
| Bounded service-timing jitter | A service-local timing tolerance class around the aligned source-release envelope. It measures admissible phase offsets while preserving service-ordered source timing. |

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| Support-source limiter guard | A local budget-preserving guard retained for over-budget support-source concentration. In the current observed-amplitude source-coupling, full-domain evolution, and aligned-envelope checks the guard is inactive; its role is to classify larger rapidity kicks that exceed row-local cone-margin budgets. |
| Causal carrier diagnostic | A radial ADM diagnostic using $v_{\pm} = -\beta \pm \alpha/\sqrt{\gamma_{ll}}$, branch-margin maps, measured cone/branch signatures, reachability masks, escape traces, and scheduled probe evolution to characterize controlled one-way carrier behavior and live-service radial escape. |
| Measured GZ-like cone signature | A measured shell/throat carrier-band feature in which a radial null branch crosses zero in the sampled ADM chart while $g_{\sigma\sigma}$ changes sign in the active support region. It is classified by reachability, radial escape, and scheduled probe gates. |
| CHL-adjacent branch-band optics | A comparison class for null-geodesic access and cone-like optical behavior in shift-driven warp geometries, following Clark, Hiscock, and Larson. In this disclosure it identifies a finite-bundle optical comparison regime for branch-band null bundles while preserving the active rail as a throat-loaded service-carrier architecture. |
| Dense finite-bundle carrier audit | A prescribed-metric null-bundle diagnostic that fans strong recovered branch-band seeds into dense fixed-time radial bundles and measures escape, recovery from both-shrinking intervals, bundle-width ratios, ray ordering, and caustic-like collapse flags. |
| Scheduled ADM probe audit | A prescribed-metric probe evolution through $\alpha(\sigma, l)$, $\beta(\sigma, l)$, $\gamma_{ll}(\sigma, l)$, and $\gamma_{\Omega\Omega}(\sigma, l)$ for packet centerline, packet edge, radial null, off-axis null, congruence, and reset-tail monitors. |
| Wake-tail restoration monitor | A reset-domain diagnostic for packet-geometry minus-branch tails after release. It records exterior clearing, live re-entry, and forward-domain closure separately from live-service carrier acceptance. |

## 7. OPERATING REGIME AND ROLE OF $V$

The service factor $V$ is the operating rating assigned to the carrying-flow sector of the reduced active-rail metric. It is a natural focus parameter because one scalar sets the plateau level of the scheduled carrying-flow service before the architecture distributes that service through support-side catch, packet rematch, radial support, and release/fade envelopes. Holding the geometric support parameters fixed while varying $V$ therefore tests how a selected rail architecture responds to a higher or lower commanded carrying-flow load.

The scheduled support-side carrying-flow factor and the packet-frame comparison factor are obtained by blending $V$ with an exit or residual service factor $v_{\mathrm{exit}}$ through the locked timing schedules:

$$U_\beta(\sigma) = v_{\mathrm{exit}} + (V - v_{\mathrm{exit}})\, C_\beta(\sigma), \tag{2}$$

$$U_{\mathrm{pkt}}(\sigma) = v_{\mathrm{exit}} + (V - v_{\mathrm{exit}})\, C_{\mathrm{pkt}}(\sigma), \tag{3}$$

where $C_\beta$ is the support-side catch schedule and $C_{\mathrm{pkt}}$ is the packet rematch schedule. In the locked-timing setting, $C_\beta$ begins with a controlled support-side lead while $C_{\mathrm{pkt}}$ is closely coupled by packet-norm safety. Thus $V$ selects the service level, while the choreography determines where and when that service level is expressed.

The carrying-flow factor enters the representative metric through the carrying-flow field $\beta$. A compact form of the support-contained carrying-flow ansatz is

$$\beta(l,\sigma) = -\,\frac{U_\beta(\sigma)\,E(\sigma)\,W(l)^{p_\beta}\,S(l,\sigma)}{Q(l,\sigma)}, \tag{4}$$

where $E$ is the release/fade envelope, $W$ is the support-shell radial window, $S$ is the packet-adjacent service envelope, and $Q = \gamma_{\Omega\Omega}$ is the throat-capacity field. The packet-frame comparison uses

$$u^l_{\mathrm{pkt}}(\sigma,l) = \frac{U_{\mathrm{pkt}}(\sigma)}{Q(l,\sigma)}, \qquad \mathcal{N}_{\mathrm{pkt}} = -\alpha^2 + G\left(u^l_{\mathrm{pkt}} + \beta\right)^2, \tag{5}$$

with packet causal safety requiring $\mathcal{N}_{\mathrm{pkt}} < 0$ on the live packet corridor. In this sense, $V$ drives the metric service along the path

$$V \to U_\beta, U_{\mathrm{pkt}} \to \beta, \mathcal{N}_{\mathrm{pkt}} \to K_{ij}, \rho_{\mathrm{ADM}}, j_l, \tag{6}$$

so a $V$-sweep probes the same architecture's carried-flow load, packet causal margin, and ADM source-placement response.

In representative diagnostics, $V = 5$ is the engineered operating embodiment. It provides the current sealed platform for ADM translation, standing-substrate subtraction, catch/rematch source-law characterization, release choreography, compact handoff shaping, source-family accounting, endpoint receiver accounting, support closure, fixed-background source-family validation, and causal-carrier acceptance. The $V = 2.5$ and $V = 10$ service-rating ladders are outside that engineered scope. They are retained as diagnostic maps for service-aware source-law design: $V = 2.5$ is live-packet safe but misses the present service-independent medium/support closure calibration, while $V = 10$ fails live packet source safety and marks the current high-service causal-margin boundary. The representative V5 operating embodiment is characterized on V5 packet safety, source placement, endpoint/support closure, and service-time rating. The beta075 receiver setting is the sealed effective receiver regime for the V5 operating point, while beta100 supplies the release-timing stress boundary and retiming anchor for stronger beta-release conditions.

### 7.1 Operational Service-Time Rating

The service factor also supplies an operational exterior-null comparison. The service-time ledger fixes exterior endpoints A and B, defines the packet departure event at A and the restored arrival event at B, and compares the reconstructed service time with an ordinary exterior null reference. In units with $c = 1$, the exterior null reference time is

$$T_{\mathrm{ext,null}} = D_{AB}. \tag{7}$$

The prepared-service time is

$$T_{\mathrm{service}} = \sigma_{\mathrm{arrive}} - \sigma_{\mathrm{depart}}, \tag{8}$$

and the two transport proxies used in the current ledger are

$$D_{\mathrm{sched}} = \int_{\sigma_{\mathrm{depart}}}^{\sigma_{\mathrm{arrive}}} U_{\mathrm{pkt}}(\sigma)\,\mathrm{d}\sigma, \tag{9}$$

$$D_{\mathrm{coord}} = \int_{\sigma_{\mathrm{depart}}}^{\sigma_{\mathrm{arrive}}} \frac{U_{\mathrm{pkt}}(\sigma)}{B(\sigma)}\,\mathrm{d}\sigma, \tag{10}$$

where $B(\sigma)$ is the centerline throat/carrying-flow capacity factor recorded in the source ledger. The advantage ratios are $D_{\mathrm{sched}}/T_{\mathrm{service}}$ and $D_{\mathrm{coord}}/T_{\mathrm{service}}$. The strict source-grid centerline control uses the modeled tube relation $l = \sigma$ and remains unity by construction.

On the $s = 15$ service-time ledger, the representative schedule has prepared-service time approximately 3.045, schedule-factor distance approximately 7.822491, and packet-coordinate distance approximately 3.755436. The corresponding operational advantage ratios are approximately 2.568962 and 1.233312, respectively. With request-triggered setup included in the service clock, the same endpoint comparison remains favorable with ratios approximately 2.487278 and 1.194097. Thus the prescribed schedule has a positive exterior-null service-time rating while packet safety, dense finite-bundle transport, source-family realization, and global causal-structure scope remain assigned to their own diagnostic gates.

### 7.2 Chronology-Governed Service Sequencing

Operational service-time advantage brings the standard superluminal chronology issue into the service-safety envelope. Everett and Roman showed the network form of the issue clearly for Krasnikov tubes: a single directed tube has one-way superluminal behavior, while a pair of tubes can be arranged as a time-machine system [3]. Shoshany and Snodgrass give a modern two-warp construction in which linked warp-drive geometries form a closed timelike geodesic [4]. In the active-rail design, the known closed-timelike-curve problem is therefore treated as a safety-engineering problem for service arming, endpoint readiness, reset timing, and network routing.

The engineering handle is monotonicity in a rail-governed time. Let $T_{\text{rail}}$ be a service chronology variable associated with a prepared rail or with a connected rail network. A permitted service edge $A \to B$ satisfies

$$T_{\text{rail}}(B_{\text{arrive}}) \geq T_{\text{rail}}(A_{\text{depart}}) + \Delta_{\text{safety}}, \tag{11}$$

where $\Delta_{\text{safety}}$ covers reset recovery, catch/release recovery, endpoint readiness, clock uncertainty, relative-motion margin, and multi-link routing margin. Return edges, cross-links, and ordinary exterior causal edges are evaluated in the same $T_{\text{rail}}$ accounting. A multi-leg service cycle is admitted after the accumulated rail-time advance satisfies

$$\sum_{e \in C} \Delta T_{\text{rail}}(e) > \Delta_{\text{cycle}}, \tag{12}$$

for the configured cycle margin $\Delta_{\text{cycle}}$.

The staged active-rail sequence supplies the control variables needed for this governor. Entry, carrier plateau, catch/release, endpoint receiver action, reset/decompression, and wake-tail restoration are distinct service states. Transport is armed during accepted rail-time service intervals and held in readiness or reset state during chronology-sensitive endpoint windows. A connected service network extends the same rule across stations and rails: all participating rails share one chronology governor or a certified translation into one global ordering. Network-scale synchronization, boosted-frame endpoint policies, and margin allocation are assigned to the subsequent safety-engineering layer.

## 8. REPRESENTATIVE REDUCED METRIC

A representative reduced metric is expressed as

$$\mathrm{d}s^2 = -\alpha(l,\sigma)^2 \mathrm{d}\sigma^2 + G(l,\sigma)\,(\mathrm{d}l + \beta(l,\sigma)\mathrm{d}\sigma)^2 + Q(l,\sigma)\mathrm{d}\Omega^2, \tag{13}$$

where $l$ is the rail coordinate and $\sigma$ is the service scheduling coordinate. The shift sector is intentionally retained in the completed-square form. This presentation displays the active-rail geometry as a Morris–Thorne-like intrinsic throat/support geometry, represented by $G$ and $Q$, with a shift-like carrying-flow sector $\beta$ applied along the rail coordinate. The four metric-side service fields are:

$$\alpha(l,\sigma) \quad \text{clock-lapse field,} \tag{14}$$
$$\beta(l,\sigma) \quad \text{carrying-flow field,} \tag{15}$$
$$G(l,\sigma)=\gamma_{ll} \quad \text{rail-stretch field,} \tag{16}$$
$$Q(l,\sigma)=\gamma_{\Omega\Omega} \quad \text{throat-capacity field.} \tag{17}$$

This notation reserves distinct symbols for metric components and support-shell overlay amplitudes.

### 8.1 General Component Form

The representative metric family can be described as a shaped baseline plus localized support-shell partners:

$$\beta(l,\sigma)=\beta_0(l,\sigma)+a_\beta W_{\rm sh}(l,\sigma), \tag{18}$$
$$\alpha(l,\sigma)=\alpha_0(l,\sigma)\exp(\kappa_\alpha W_{\rm sh}(l,\sigma)), \tag{19}$$
$$G(l,\sigma)=G_0(l,\sigma)\exp(\kappa_G W_{\rm sh}(l,\sigma)), \tag{20}$$
$$Q(l,\sigma)=Q_0(l,\sigma)\exp(\kappa_Q W_{\rm sh}(l,\sigma)). \tag{21}$$

The baseline fields $\alpha_0, \beta_0, G_0, Q_0$ include the prepared throat, angular jacket, radial support-edge softening, lapse cushion, entry service gate, shaped catch/rematch timing, service release, packet-edge beta rematch, packet-local radial support shaping, compact entry/catch/handoff support, and packet-frame comparison structure. The localized partners are support-shell metric overlays. The continuous partner family includes carrying-flow ($a_\beta$), clock-lapse ($\kappa_\alpha$), rail-stretch ($\kappa_G$), and throat-capacity ($\kappa_Q$). Support-shell source-ledger accounting assigns carrying-flow plus clock-lapse as the default partner pair, rail-stretch as a peak-shaping comparator, and same-window throat-capacity as neutral in the default setting. The representative V5 operating embodiment uses release choreography, a blended trailing-edge packet beta-rematch sleeve, a two-feature packet-local radial support profile, and compact pressure-aware handoff support to preserve live packet safety and control demanded-source placement through the catch/release handoff.

The support-shell window is an annular, timed, packet-excluding window:

$$W_{\rm sh}(l,\sigma)=R_{\rm sh}(|l|)\,T_{\rm sh}(\sigma;\lambda_c,\tau_c)\,P_{\rm excl}(l,\sigma), \tag{22}$$

where $R_{\rm sh}$ selects support-edge infrastructure, $T_{\rm sh}$ places the control action relative to catch/rematch, and $P_{\rm excl}$ suppresses direct live-packet overlap. In the representative implementation,

$$0.65\,R_{\rm th}\le|l|\le 1.20\,R_{\rm th} \tag{23}$$

is the nominal support-shell band, $\lambda_c$ is the support-shell catch lead, and $\tau_c$ is the temporal width of the support-shell timing envelope.

The preferred V5 support-shell source-ledger comparator uses a compact raised-cosine annulus centered in the nominal support-shell band:

$$R_{\rm sh}(r)=\frac{1}{2}\left[1+\cos\!\left(\pi\frac{|r-r_c|}{h_r}\right)\right], \qquad |r-r_c|\le h_r, \tag{24}$$
$$R_{\rm sh}(r)=0, \qquad |r-r_c|>h_r, \tag{25}$$

where $r=|l|$, $r_c$ is the midpoint of the active support-shell band, and the preferred half-width is approximately $h_r=0.48125$ for $R_{\rm th}=1.75$ with the above multipliers. Gaussian and smooth-box annular windows are retained as comparator shapes, and the raised-cosine annulus gives the cleanest source-ledger behavior in matched-strength V5 accounting.

The preferred timing envelope is Gaussian in the service coordinate and normalized against the active catch/rematch edge:

$$T_{\mathrm{sh}}(\sigma;\lambda_c,\tau_c)=\frac{\exp[-(\sigma-\sigma_c)^2/(2\tau_c^2)]}{\max\left(\exp[-(\sigma_*-\sigma_c)^2/(2\tau_c^2)],\epsilon_T\right)},\qquad \sigma_c=\sigma_{\mathrm{catch}}-\lambda_c, \tag{26}$$

where $\sigma_*$ is the closest point in the active catch/rematch band to $\sigma_c$. Minimum-jerk and raised-cosine temporal pulses are useful comparators. The representative timing band is $\tau_c = 0.30$–$0.35$ with $\lambda_c = 1.45$–$1.55$ for the V5 support-shell source-ledger family.

For matched-strength shape comparison, the support-shell carrying-flow overlay is parameterized by the peak shell perturbation

$$\Delta\beta_{\mathrm{sh,max}}=\max_{l,\sigma}\left|a_\beta W_{\mathrm{sh}}(l,\sigma)\right|. \tag{27}$$

This scalar makes radial and temporal shapes comparable at equal effective peak support-shell strength. The preferred V5 operating band is $\Delta\beta_{\mathrm{sh,max}} \approx 0.20$–$0.25$, with $\Delta\beta_{\mathrm{sh,max}} \approx 0.30$–$0.35$ used as an edge-stress range for point-peak and shell-throat concentration checks.

### 8.2 Release Choreography, Compact Handoff, and Packet-Edge Rematch

The representative V5 operating embodiment adds a release-choreography layer to the support-shell infrastructure. The release law is a coordinated handoff in which the packet is first caught/rematched, briefly held in a matched state, released through a finite smooth beta fade, and protected by lapse/carve support until packet-edge mismatch has been transferred into the designated handoff windows.

A representative release envelope is written as

$$W_{\mathrm{rel}}(\sigma)=\mathcal{S}_5\left(\frac{\sigma-\sigma_{\mathrm{hold}}}{\Delta\sigma_\beta}\right),\qquad \mathcal{S}_5(x)=10x^3-15x^4+6x^5,\quad 0\le x\le 1, \tag{28}$$

with constant matched hold before the finite fade and saturated values outside the fade interval. Higher-smoothness profiles remain comparator choices, but the present preferred release setting uses a minimum-jerk profile because its endpoint derivatives vanish and because it keeps the release law interpretable.

The compact handoff layer expresses the same release principle spatially. It uses separate endpoint-smooth windows for entry containment, catch/rematch bearing, and trailing-edge handoff smoothing:

$$W_{\mathrm{ch}}(l,\sigma)=g_eW_e(l,\sigma)+g_cW_c(l,\sigma)+g_hW_h(l,\sigma), \tag{29}$$

where $W_e$ supplies broad live-entry pressure containment, $W_c$ supplies catch/rematch bearing, and $W_h$ supplies outer packet-edge derivative smoothing. A representative high-resolution V5 compact handoff uses a seventh-order smoothstep transition, an entry containment strength near 0.75, catch strength near 0.15, edge strength near 0.16, a broad edge width multiplier near 7.2, and an entry service gate beginning near $\sigma = -1.40$. This structure gives explicit control over radial pressure through the entry window while assigning angular/current and radial-null derivative relief to the catch and edge windows.

The entry service gate is a source-accounting boundary that represents setup physics in the full infrastructure ledger and begins live packet scoring once the packet has entered the supported corridor. In the representative high-resolution V5 ledger on the extended $101 \times 151$ domain, the entry-gated compact handoff has zero positive live packet-norm samples and a maximum live packet norm of approximately $-1.2424$.

The packet-edge beta rematch acts on the local mismatch

$$v_{\mathrm{coord}}+\beta_{\mathrm{pre}}, \tag{30}$$

where $\beta_{\rm pre}$ denotes the carrying-flow field before packet-local beta rematch. A compact notation for the packet-local correction is

$$\delta\beta_{\rm pkt}(l,\sigma) = -g_\beta\, W_{\beta,{\rm pkt}}(l,\sigma)\ (v_{\rm coord} + \beta_{\rm pre}(l,\sigma))\,. \tag{31}$$

The preferred $W_{\beta,{\rm pkt}}$ is a trailing-edge sleeve: weak in the packet center, strongest near the inner/behind side of the packet edge, and temporally active through the live catch/rematch corridor. The shaped sleeve is combined with a weak filled-core floor using a smooth blended union,

$$W_{\beta,{\rm pkt}} = W_{\rm sleeve} + W_{\rm floor}\,(1 - W_{\rm sleeve})\,, \tag{32}$$

which gives a single continuous packet-edge correction profile. The high-resolution V5 operating embodiment uses a beta-rematch gain of about 1.8, center floor 0.60, inner/outer sleeve radii approximately $0.75\,R_{\rm pass}$ and $1.10\,R_{\rm pass}$, and widened radial transition width multiplier 1.6.

The blended floor is paired with sufficient trailing-edge sleeve width. In the V5 operating family, the wider sleeve preserves packet gates under refinement and substantially lowers point peaks. The compact handoff layer then broadens the same principle across entry, catch, and release by assigning pressure containment and derivative smoothing to separate smooth windows.

### 8.3 Packet-Local Radial Support Profile

The packet-local radial support profile modifies $G = \gamma_{ll}$ through a smooth multiplicative factor:

$$G(l,\sigma) = G_{\rm pre}(l,\sigma)\exp(g_c W_c(l,\sigma) + g_r W_r(l,\sigma) + g_s W_s(l,\sigma))\,, \tag{33}$$

where $G_{\rm pre}$ denotes the radial metric before packet-local radial support, $W_c$ is a filled packet/support core window, $W_r$ is a narrow annular catch ring, and $W_s$ is a weaker annular skirt. In the representative V5 setting,

$$g_c = +0.05, \qquad W_c: \text{ entry/catch/release schedule, radius } 1.3R_{\rm pass},\ \text{width } 1.8w_{\rm pass}, \tag{34}$$

$$g_r = +0.12, \qquad W_r: \text{ catch-only annulus, outer radius } 2.4R_{\rm pass},\ \text{width } 2.4w_{\rm pass}, \tag{35}$$

$$g_s = +0.05, \qquad W_s: \text{ catch-only annulus, inner/outer radii } 2.4R_{\rm pass}, 2.6R_{\rm pass},\ \text{width } 3.2w_{\rm pass}. \tag{36}$$

The core/ring/skirt structure separates two radial support functions. The narrow ring provides the principal radial-null and angular-pressure localization benefit. The weak skirt provides a smaller smoothing correction to live $p_l$ and $p_\Omega$ while preserving the main point-peak relief. High-resolution tests show that this two-feature radial profile keeps all live packet-norm samples safely negative and lowers the radial-null point peak and angular-pressure burden relative to the release-only comparison family.

### 8.4 Representative Settings

Representative parameter values are listed in Table 2. The values define the working design point and its comparator settings.

Table 2: Representative active-rail parameters and functional implementation decisions.

| **Parameter** | **Representative value or status** |
|---|---|
| Engineered service factor | $V = 5$ |
| Lower service-rating diagnostic | $V = 2.5$, outside current engineered scope; live-packet safe but medium/support closure calibration fails |

| **Parameter** | **Representative value or status** |
|---|---|
| High service-rating diagnostic | $V = 10$, outside current engineered scope; live packet source safety fails and additional causal-margin engineering is required |
| Packet radius | $R_{\text{pass}} = 0.35$ |
| Support radius | $R_{\text{th}} = 1.75$ |
| Angular jacket radius | $R_{\Omega} = 1.75$ |
| Radial support width | $w_{\text{th}} = 0.569$ |
| Packet transition width | $w_{\text{pass}} = 0.06$ |
| Angular jacket width and amplitude | $w_{\Omega} = 1.4$, $a_{\Omega} = 0.20$ |
| Lapse cushion strength | $\eta_N = 2.00$ |
| Catch/rematch profile | matched-hold release choreography with minimum-jerk beta fade |
| Reduced support-shell routing overlay | $a_\beta = +10^{-7}$ as the small reduced-routing target; high-amplitude release-choreography comparators use service shell amplitude 0.5 |
| Support-shell radial band | $0.65\, R_{\text{th}} \le \lvert l \rvert \le 1.20\, R_{\text{th}}$ |
| Support-shell radial shape status | raised-cosine annulus as the cleanest matched-strength support-shell source comparator; release-choreography accounting uses the smooth-box support-shell infrastructure knob |
| Selected 4D support-shell strength | $\Delta\beta_{\text{sh,max}} \approx 0.20$–$0.25$ for support-shell source comparators |
| Selected 4D timing band | $\lambda_c = 1.45$–$1.55$, $\tau_c = 0.30$–$0.35$ |
| Clock-lapse partner | $\kappa_\alpha / a_\beta \approx 0.375$ for the selected V5 support-shell family |
| Rail-stretch partner | $\kappa_G / a_\beta = 0$ by default; positive rail-stretch as a comparator for peak shaping |
| Throat-capacity partner | $\kappa_Q / a_\beta = 0$ by default in the selected support-shell setting; alternate timing and separate intrinsic windows remain comparator knobs |
| Packet carve/lapse support | packet exclusion 0.90, annular shoulder 0.18, packet-lapse log gain 1.0 with entry/catch/release timing |
| Packet beta-rematch sleeve | trailing-edge shape, gain 1.8, smooth blended floor 0.60, inner/outer radii $0.75\, R_{\text{pass}}$ and $1.10\, R_{\text{pass}}$; the conservative finite-bundle carrier lead uses sleeve transition multiplier 6.0 and temporal multiplier 1.5, with multiplier 8.0 and temporal multiplier 2.0 as the wider optical-relief bracket |
| Packet-local radial support profile | filled core gain +0.05, annular catch ring gain +0.12 at radius/width 2.4/2.4, and weak annular skirt gain +0.05 with inner/outer/width 2.4/2.6/3.2 |
| Compact handoff layer | seventh-order smoothstep entry/catch/edge windows, entry service gate near $\sigma = -1.40$, broad edge transition multiplier near 7.2, and separated entry pressure containment versus catch/edge derivative smoothing |

| Parameter | Representative value or status |
|---|---|
| Semilocal null-accumulation diagnostic | Gaussian or equivalent smooth radial-null averaging on selected live, handoff, endpoint, and source-sector windows, with affine-reparameterized radial-null benchmark comparisons used as acceptance diagnostics |
| Source-family grammar | $S_0 + J + R + C_{\text{live}} + G + D_H$, assigning the non-live radial body to a constant-flux string-cloud backbone, endpoint residuals to a coupled junction layer, live burden to handoff trim, angular burden to capacity support, and non-live current to relaxation sectors |
| Representative endpoint receiver | sealed beta075 $p003\_mid$ negative-$l$ beta-memory receiver for the engineered $V = 5$ operating point, with localized angular endpoint support and dense live-clean source-sector accounting |
| Endpoint/source tensor closure | sealed beta075 repaired-lead endpoint/source components close as a covariant finite-difference tensor before support exchange is added |
| Support stroke/stress closure | phase-aware support stroke/stress-potential sector supplies the divergence-form support exchange current; the sealed total-closure reference uses the $24 \times 14$ support-stress basis |
| Compact support-stress bracket | $22 \times 13$ support-stress basis is retained as a compactness bracket because aggregate and localization gates pass but the dense local active total-closure gate is tighter than the compact basis |
| Reduced principal-symbol layer | fixed-background endpoint/support characteristic diagnostic on $U = (h, \psi, \chi_\Omega, \pi_\Omega, \Phi_{\text{support}}, \Pi_{\text{support}}, n_l)$; promoted dense $24 \times 14$ run has zero hard gate-crossing rows, complete in-cone eigenbases, and minimum cone margin $7.881 \times 10^{-5}$ |
| Bounded rapidity transport | bounded transport rapidity is the lead reduced heat/current variable; raw $+10^{-4}$ heat-ratio stress crosses one reset/support-edge local gate, while the bounded-rapidity $+10^{-4}$ stress clears the hard gates with minimum cone margin $2.252 \times 10^{-5}$ |
| Rapidity budget and advection guard | tightest observed row is reset-decompression/support-edge row 4605; observed $\Delta\psi = 0.626342$ uses 0.286844 of its budget, monotone inward/outward advection keeps it within budget, and the local limiter guard is inactive at observed amplitude |
| Beta100 timing anchor | beta100 stress operation uses the same receiver grammar with service-specific retiming; the post-release 1.5 anchor gives positive selected, current, and angular transfer accounting on the stress grid |
| Causal-carrier acceptance | measured GZ-like branch structure is characterized in the shell/throat carrier band and accepted through zero service/carry entry-to-packet reachability, branch-band radial escape, scheduled ADM probes, dense finite-bundle carrier transport, exterior-null service-time rating, and reset-domain wake-tail monitors |
| Service-time advantage rating | representative $s = 15$ exterior-null comparison gives prepared-service advantage ratios about 2.568962 under the schedule-factor proxy and about 1.233312 under the packet-coordinate proxy |

| Parameter | Representative value or status |
|---|---|
| Representative V5 status | high-resolution packet-safe, with structured burden separated between live entry/handoff rows, semilocal radial-null exposure, non-live catch/rematch radial-null rows, mitigated CHL-adjacent branch-band optics, favorable exterior-null service-time rating, sealed beta075 endpoint source-medium closure, total endpoint/support exchange closure at finite-difference level, and necessary reduced fixed-background support-sector principal-symbol and rapidity-transport gates cleared at observed amplitude |

## 9. SERVICE SEQUENCE

A representative service cycle proceeds through the following phases:

$$\begin{gathered}\text{prepared substrate} \to \text{ready support plant} \to \text{packet entry} \to \text{locked catch/rematch} \\ \to \text{support-contained carrying-flow handoff} \to \text{packet release} \to \text{infrastructure reset.}\end{gathered} \tag{37}$$

Passenger-facing accounting is assigned to packet entry, catch/rematch, carried service, release, and selected live buffer windows. System-level ledgers may include reset and decompression through an explicitly selected infrastructure diagnostic window.

## 10. ADM TRANSLATION

The metric of Eq. (13) is expressed in ADM form through

$$\gamma_{ij} = \mathrm{diag}\left(G, Q, Q\sin^2\theta\right), \qquad \beta^i = (\beta, 0, 0), \qquad \beta_i = (G\beta, 0, 0). \tag{38}$$

The unit normal to a $\sigma =$ constant slice is

$$n^\mu = \left(\frac{1}{\alpha}, -\frac{\beta}{\alpha}, 0, 0\right), \qquad n_\mu = (-\alpha, 0, 0, 0). \tag{39}$$

Using dot notation for $\partial_\sigma$ and prime notation for $\partial_l$, and adopting the convention

$$K_{ij} = \frac{1}{2\alpha}\left(D_i\beta_j + D_j\beta_i - \partial_\sigma\gamma_{ij}\right), \tag{40}$$

the independent extrinsic-curvature components are

$$K_{ll} = \frac{1}{2\alpha}\left(2G\beta' + G'\beta - \dot{G}\right), \tag{41}$$

$$K_{\theta\theta} = \frac{1}{2\alpha}\left(\beta Q' - \dot{Q}\right), \tag{42}$$

$$K_{\phi\phi} = K_{\theta\theta}\sin^2\theta. \tag{43}$$

The mixed eigenvalue-like quantities are

$$k_l = K^l{}_l = \frac{1}{2\alpha}\left(2\beta' + \beta\frac{G'}{G} - \frac{\dot{G}}{G}\right), \tag{44}$$

$$k_\Omega = K^\theta{}_\theta = K^\phi{}_\phi = \frac{1}{2\alpha}\left(\beta\frac{Q'}{Q} - \frac{\dot{Q}}{Q}\right). \tag{45}$$

Thus

$$K = k_l + 2k_\Omega, \qquad K_{ij}K^{ij} = k_l^2 + 2k_\Omega^2. \tag{46}$$

For spatial line element $\mathrm{d}l_3^2 = G\mathrm{d}l^2 + Q\mathrm{d}\Omega^2$, the scalar curvature is

$$^{(3)}R = -\frac{2Q''}{GQ} + \frac{(Q')^2}{2GQ^2} + \frac{G'Q'}{G^2Q} + \frac{2}{Q}. \tag{47}$$

The Hamiltonian and radial momentum constraints produce the ADM ledger fields

$$16\pi\rho_{\mathrm{ADM}} = {}^{(3)}R + K^2 - K_{ij}K^{ij} = {}^{(3)}R + 4k_l k_\Omega + 2k_\Omega^2, \tag{48}$$

$$8\pi j_l = -2k'_\Omega + \frac{Q'}{Q}(k_l - k_\Omega), \tag{49}$$

$$j_\theta = 0, \qquad j_\phi = 0. \tag{50}$$

## 11. SOURCE-PLACEMENT LEDGER

The source-placement ledger assigns demanded source roles to system components. The ledger is used for engineering localization, source-family accounting, and representative-design acceptance.

| Source role | Engineering assignment | Primary diagnostic |
|---|---|---|
| Standing support substrate | Prepared throat/core, support shell, baseline clock-lapse, and angular jacket | baseline $\rho_{\mathrm{ADM}}$ and pressure channels |
| Radial support edge | Radial softening and support-shell gradient control | $p_l$, edge gradients |
| Throat-capacity/angular jacket | Angular and throat support capacity | $p_\Omega$, $T_{kk}$ side effects |
| Clock-lapse cushion | Causal-margin and timing support | packet norm margin, $T_{kk}$ response |
| Carrying-flow field | Support-contained scheduled service flow | $\Delta j_l$, packet norm |
| Support-shell overlay | Annular support infrastructure timing component | catch/support $\Delta j_l$ fraction, 4D source ratios |
| Catch/rematch absorber | Inner packet-edge handoff absorber | $\Delta j_l$, $T_{kk}$ residual |
| Release choreography | Matched hold, derivative-limited beta fade, and coordinated lapse/carve relaxation | packet norm margin, release derivative maxima, live high-burden channel exposure |
| Compact handoff layer | Endpoint-smooth entry containment, catch bearing, and edge smoothing | live $p_l$, $j_l$, $p_\Omega$, semilocal $T_{kk}$ |
| Entry service gate | Service boundary that separates setup support from live packet scoring | live packet norm, live source-role closure |
| Trailing-edge beta-rematch sleeve | Smooth packet-edge correction for local $(v_{\mathrm{coord}} + \beta)$ mismatch and branch-band carrier-collar width preservation | packet-norm safety, live $T_{kk}$ fraction, point peaks, dense bundle width |

| | | |
|---|---|---|
| Packet-local radial support profile | Mild $\gamma_{ll}$ core stretch plus annular catch ring and weak skirt | $T_{kk}$, $p_l$, $p_\Omega$, radial metric derivative diagnostics |
| Semilocal null-accumulation ledger | Finite-window radial-null averaging across live packet, handoff, endpoint, and source-sector regions | finite-window $T_{kk}$ floor margin and negative-part integral |
| Finite-domain radial ANEC alignment | Longest available affine radial-null trace integrals used to locate averaged-null balance across completed and residual source sectors | net $\int T_{kk}\,d\lambda$, positive/negative parts, sector attribution, domain coverage |
| Constant-flux radial string-cloud backbone | Non-live radial-tension support scaffold for the dominant A/B/I radial infrastructure sector | $\rho$, $p_l$, areal flux proxy, live fraction |
| Endpoint/junction layer | Coupled support-edge and reset/decompression endpoint source target carrying residual radial trim, angular support, and current relaxation | selected-null deficit, current, $p_\Omega$, finite-width closure, regulated medium response |
| Beta-memory endpoint receiver | Negative-$l$ support-edge receiver coupled to beta-release transfer memory | endpoint transfer accounting, live gate, source-sector closure |
| Endpoint current regulator and medium | Non-live regulated anisotropic heat/current endpoint medium with a small radial-pair regulator and internal angular response | radial-block class, heat-flux margin, angular closure, live exclusion |
| Support stroke/stress-potential sector | Divergence-form support exchange completion paired with the endpoint/source tensor | total conservation, localization, support-tail audit, live exclusion |
| Causal carrier layer | Controlled one-way radial carrier structure generated by $\alpha$, $\beta$, and $\gamma_{ll}$ | branch margins, entry reachability, radial escape, probe evolution |
| Reset/decompression infrastructure | Post-service restoration of support state | infrastructure ledger |

The reduced ADM diagnostics indicate that the selected small support-shell overlay places its incremental $j_l$ demand overwhelmingly in catch/rematch support infrastructure while keeping packet $\Delta\rho$ effectively quiet. The 4D demanded-source diagnostics sharpen that result: the same component is packet-safe and controllable as a continuous metric overlay, while larger single-channel carrying-flow amplitude identifies the radial-null and current channels assigned to coupled support. The release-choreography, radial-support, compact-handoff, and entry-gate ledgers further sharpen the design. V5 packet safety is maintained through a smoother blended trailing-edge beta rematch, two-feature packet-local radial support, compact pressure-aware entry/catch/edge shaping, and live scoring that begins inside the supported entry corridor. The demanded-source burden is structured: live angular/current peaks are assigned to packet-in-support rows, semilocal radial-null exposure is measured by finite-window null-accumulation diagnostics, non-live radial-null plus radial-pressure burdens are carried by core-throat, support-edge, and endpoint infrastructure, and the beta075 endpoint package is represented by a regulated non-live heat/current medium paired with a phase-aware support stroke/stress-potential exchange completion.

### 11.1 Semilocal Null-Accumulation Source Ledger

The source-placement ledger includes semilocal radial-null accounting for comparing pointwise $T_{kk}$ concentration with finite-window exposure. The diagnostic follows quantum-motivated finite-window null-energy logic in a prescribed-metric engineering ledger [7]. For a diagnostic center $(l_c, \sigma_c)$, define

$$\langle T_{kk} \rangle_{W_{\rm SN}} = \frac{\sum_i W_{\rm SN}(l_i, \sigma_i; l_c, \sigma_c)\, T_{kk}(l_i, \sigma_i)}{\sum_i W_{\rm SN}(l_i, \sigma_i; l_c, \sigma_c)}, \tag{51}$$

where $W_{\rm SN}$ is a compact Gaussian or equivalent smooth averaging window on the selected live-packet, handoff, endpoint, or source-sector scope. A representative benchmark compares this finite-window value with

$$T_{kk,\rm floor} = -\frac{8\pi B}{\tau^2}, \qquad B = \frac{1}{32\pi}, \tag{52}$$

so that $\tau = 1$ gives $T_{kk,\rm floor} = -0.25$ in ledger units. The compact handoff family improves the finite-window radial-null margin on the live catch/rematch packet window while also reducing broad angular-pressure exposure. The dense entry-gated source-sector closure, intermediate source targets, and endpoint/junction receiver ledgers retain positive benchmark margin on the finite windows used for representative acceptance.

For accepted radial-null SNEC accounting, the window coordinate is affine-reparameterized along the sampled radial null branch. With coordinate-time tangent

$$K^\mu = (1, v_\pm), \tag{53}$$

the diagnostic evaluates

$$K^\nu \nabla_\nu K^\mu = \kappa K^\mu + \mathcal{R}^\mu_\perp, \tag{54}$$

where $\kappa$ is the measured non-affinity and $\mathcal{R}^\mu_\perp$ is the remaining radial geodesic residual monitor. The affine weight is integrated from $\kappa$ and normalized at the window center. In the repaired beta075 *rematch_w6_t1p5* package, the full-stride affine-reparameterized radial-null screen evaluates 137,064 windows at affine widths 0.5, 1.0, and 2.0 in both component-sector and intermediate residual-inclusive modes, with zero all-window and zero scoreable benchmark-floor violations. The tightest scoreable broad-window margin is approximately 0.051880 and remains non-live. This diagnostic expresses source cost at the finite engineering scale of the packet, handoff, endpoint, and source-sector structures and complements the point-peak ledger.

The finite-domain radial ANEC companion connects that finite-window SNEC result to the Barceló–Visser energy-condition comparison class, where warp/throat-like support pays an averaged-null exotic-source price [6]. It integrates $T_{kk}\, d\lambda$ along the longest available in-domain affine radial-null traces seeded from uniform, worst-window, endpoint/support, and live-handoff rows. On the repaired beta075 *rematch_w6_t1p5* package, the explicit non-live sector package is close to finite-domain radial-ANEC balance once the source-completion residual and live handoff trim are separated: removing the closure residual makes the minus branch nearly balanced, and removing both closure residual and live handoff trim leaves worst remaining branch integrals about $-0.004643$ and $-0.009361$. The demanded geometric total records the remaining endpoint/support completion load, with median branch integrals about $-0.067$ and dominant attribution to `sector_closure_residual`. In rail terms, the averaged-null cost is an endpoint/support source-plant target while the live carrier corridor remains protected by the repaired SNEC-clean package. Because the residual-stripped traces approach balance on the same carrier geometry, the data points toward source-family refinement as the route to a finite-domain radial-ANEC-clean rail. The source-family question is whether the endpoint/support medium can supply that localized cost with controlled field amplitudes and gradients, preserving the rail's advantage: a passenger/live corridor carried by managed support architecture rather than by the main exotic source layer.

The associated boundedness ledger records effective-model support budgets. At the 1.10 regulator pad, its archived endpoint rows have zero live regulator rows, zero reported post-regulator Type-IV rows, and zero reported superluminal or undefined boosts; the symbol and energy ledgers record zero hard failures. These classifier-dependent physical interpretations require corrected reevaluation. The operator and exchange-fit constants retain their stated scope: minimum transport/cone margin about $7.9\times10^{-5}$, stored Type-I margin about $4.0\times10^{-9}$, local regulator/source ratio about 1.0527, local power-channel shape about 0.6090, and exchange-fit headroom about 0.005433. A physical specialization supplies the reservoir tensor, stored enthalpy response, and complete radial/angular source balance while preserving live-corridor protection.

## 12. CATCH-EDGE CONTROL LAW

Let

$$\xi = l - s(\sigma) \tag{55}$$

be a packet-centered radial coordinate. A preferred absorber amplitude is a compact kinematic handoff law multiplied by an inner-edge envelope:

$$A_{\text{catch}}(\xi,\sigma) = W_{\text{catch}}(\sigma)\left[a\,H_{\text{flow}}(\xi,\sigma) + b\,B_{\text{inner}}(\xi)\right], \tag{56}$$

where $W_{\text{catch}}$ is a locked timing window, $H_{\text{flow}}$ is a smooth carrying-flow handoff or flow-mismatch measure, and $B_{\text{inner}}$ is a compact bump centered near $\xi = -R_{\text{pass}}$. A residual-responsive limiter may bound the amplitude:

$$A_{\text{catch}} \le C\left([R_{T_{kk}}]_+\right), \tag{57}$$

where $C$ denotes a smooth cap and $R_{T_{kk}}$ denotes the predicted radial-null residual. The primary control law is kinematic; the residual-responsive term supplies protective limiting and diagnostic alignment.

## 13. STANDING SUBSTRATE AND SOURCE-LEDGER RESULTS

Standing substrate subtraction isolates service-induced ADM demand from prepared infrastructure demand. For a live service field $F_{\text{live}}$ and a time-matched prepared carrying-flow-off substrate field $F_{\text{sub}}$, define

$$\Delta\rho_{\text{ADM}} = \rho_{\text{ADM,live}} - \rho_{\text{ADM,sub}}, \tag{58}$$

$$\Delta j_l = j_{l,\text{live}} - j_{l,\text{sub}}. \tag{59}$$

This diagnostic separates baseline support curvature from active service increments. The heavier demanded-source ledger evaluates

$$T^{\text{dem}}_{\mu\nu} = \frac{1}{8\pi}G_{\mu\nu}[g] \tag{60}$$

from the prescribed four-dimensional metric and records radial null demand, radial pressure, angular pressure, radial current, and packet-comoving density.

The following table retains the archived results with their original grids and normalizations. Source-role coverage measures demanded-channel assignment; endpoint tensor identity measures the fitted medium; support closure measures the complementary exchange-current fit. Effective characteristic and energy margins belong to their specified operators. Rest-frame-dependent columns require the corrected classifier for physical admissibility.

Table 4: Representative reduced ADM and 4D demanded-source evidence.

| **Diagnostic layer** | **Representative result** |
|---|---|
| $V = 5$ exact ADM and substrate subtraction | Live packet norm samples are safe. The packet contribution to subtracted $\Delta\rho$ is about 0.000295% of live-packet $\rho$, while live-packet $\Delta j_l$ is about 11.75% of live-packet $j_l$. This is the engineered operating embodiment. |
| $V = 2.5$ service-rating diagnostic | Fresh same-quality Stage II source-family ladder remains live-packet safe and covariantly coherent, but misses the service-independent endpoint medium/support closure calibration. This is outside the current engineered V5 scope and motivates an explicit V-aware closure law. |
| $V = 10$ service-rating diagnostic | Fresh same-quality Stage II source-family ladder fails live packet source safety with positive live packet-norm samples. Downstream covariant identity/localization remains coherent, but the case is outside the current engineered V5 scope and requires additional high-service causal-margin engineering. |
| Reduced support-shell routing target | At $V = 5$, the selected $a_\beta = +10^{-7}$ target gives catch/support incremental $\Delta j_l$ fraction 0.9928425, packet incremental $\Delta j_l$ fraction $3.19 \times 10^{-7}$, packet incremental $\Delta\rho = 5.44 \times 10^{-16}$, and zero packet norm violations. |
| Small-target historical $V = 10$ packet safety | Earlier reduced small-target support-shell checks remain useful as low-amplitude channel-placement evidence, but they are not the current beta075 Stage II service-rating seal. The current full beta075 V10 ladder fails live packet source safety. |
| Single-channel continuous carrying-flow overlay | High-amplitude continuous carrying-flow overlays are packet-safe at $V = 5$ and identify radial-null and current burden channels. At $a_\beta = +0.5$ on the high-resolution check, max total burden ratio is about 1.090805, radial-null ratio about 1.056153, and radial-current ratio about 1.040020. |
| Timed coupled support-shell refinement | High-resolution timing checks select a delayed, narrower support-shell event. With positive carrying-flow overlay, $\lambda_c = 1.45$–$1.55$, $\tau_c = 0.30$, $\kappa_\alpha/a_\beta = 0.375$–$0.5$, and $\kappa_G/a_\beta = 0$, max total burden improves to about 1.053–1.054 in the smooth-box support-shell family. |
| Throat-capacity accounting | Same-window throat-capacity coupling gives its best source-objective and aggregate behavior at $\kappa_Q/a_\beta = 0$ in the V5 support-shell setting. Negative ratios offer small localized tradeoffs with peak and radial-current penalties, while positive ratios raise radial-null burden. |
| Matched-strength shape selection | At equal peak $\Delta\beta_{\mathrm{sh,max}} = 0.25$, the raised-cosine annulus outperforms smooth-box and Gaussian annular profiles. The best matched-strength case gives source-objective score 0.803284, max burden ratio 1.013261, live packet-norm drift $3.09 \times 10^{-8}$, radial-null ratio 1.007942, radial-current ratio 1.002684, and angular-pressure ratio 1.013261. |

| **Diagnostic layer** | **Representative result** |
|---|---|
| Strength robustness ladder | The selected raised-cosine annulus is packet-quiet over $\Delta\beta_{\mathrm{sh,max}} =$ 0.15–0.35. The preferred operating band is 0.20–0.25; higher targets are useful edge-stress cases for point-peak and shell-throat concentration diagnostics. |
| Release choreography | A V5 matched-hold release law with minimum-jerk beta fade, packet carve/lapse support, and a trailing-edge beta rematch preserves packet safety on the representative grid with all live packet-norm samples safely negative and zero high-burden live top points. |
| Blended trailing-edge beta rematch | A smooth blended packet-center floor with a widened trailing-edge sleeve survives $81 \times 109$ with all live packet-norm samples safely negative, zero high-burden live top points, max packet norm about $-5.455516$, live $T_{kk}$ fraction about 0.024255, live $p_l$ fraction about 0.006986, max total burden ratio about 2.035942, and max point peak about 8.627056. The finite-bundle carrier lead widens the same trailing-edge rematch family for branch-band bundle preservation. |
| Two-feature radial support profile | Adding a mild $\gamma_{ll}$ core stretch, narrow annular catch ring, and weak annular skirt preserves packet safety and keeps the main point-peak relief. At high resolution it lowers the radial-null point peak by about 4.0% and lowers live angular-pressure burden by about 2.45% relative to the corresponding blended release ledger. |
| V5 algebra ledger | On the full-range $81 \times 109$ algebra ledger, live packet safety is hard-bounded with zero positive live packet-norm points and max live packet norm about $-5.455491$. Top live angular/current rows lie in entry-precatch packet-in-support samples ahead of the radial core/ring/skirt window onset, while top non-live radial-null rows lie in catch/rematch core-throat and support-edge samples. |
| Compact handoff law | At $61 \times 83$, the compact wide-edge handoff lowers live $T_{kk}$ by about 10.8%, live $j_l$ by about 10.2%, and live $p_\Omega$ by about 35.9% relative to the split-carve reference while preserving packet safety. The same compact handoff reduces channel-cause radial-metric, lapse-curvature, beta-gradient, and radial second-derivative scores in the live-badness ledger. |
| Semilocal compact handoff | On the live catch/rematch packet scope with $\tau = 1$, the split-carve reference gives finite-window $T_{kk,\min} \approx -0.2317$ against the $-0.25$ benchmark floor. The compact wide-edge handoff gives finite-window $T_{kk,\min} \approx -0.1974$, increasing the finite-window margin from about 0.0183 to about 0.0526 while reducing the negative-part integral. |
| Entry service gate and compact handoff | On the extended $101 \times 151$ domain, the representative entry-gated compact handoff has zero positive live packet-norm samples and max live packet norm about $-1.2424$. Live channel assignment closes with zero live residual fraction for radial-null, radial-pressure, radial-current, and angular-pressure burdens. |

| **Diagnostic layer** | **Representative result** |
|---|---|
| Pressure-aware entry/handoff shaping | Stronger broad entry containment lowers live $T_{kk}$ and live $p_l$, while separated entry, catch, and edge functions assign angular/current and point-peak channels to distinct source roles. The representative pressure-aware handoff separates entry pressure containment from catch/edge derivative smoothing so those channels can be tuned independently. |
| Full-grid source decomposition | The entry-gated compact wide4 source picture is support-plant dominated. The integrated live fractions are about 2.889% for radial-null burden, 0.787% for radial pressure, 14.706% for angular pressure, and 11.976% for current, with zero positive live packet-norm samples. |
| Dense source-sector closure | On the dense entry-gated sector closure, constrained source sectors assign about 98.659% of radial-null burden, 98.716% of radial-pressure burden, and 91.315% of angular-pressure burden into explicit source roles. The promoted non-live current-relaxation sector carries the remaining small current burden with zero live burden. |
| Source-family target split | The principal source target is the A/B/I non-live radial infrastructure family. In representative source-target accounting, A/B/I carries about 94.03% of the target burden, live handoff trim about 2.88%, angular capacity about 1.50%, and current control about 0.11%. |
| Constant-flux radial string-cloud backbone | The A/B/I body is close to an NEC-saturating radial-tension scaffold with $p_l/\rho \approx -0.999288$, zero live scaffold fraction, and selected-null deficit about 1.338% of the string-like scaffold. The areal string-flux proxy has weighted relative mean deviation about 0.0904%. |
| Endpoint-J source freeze | After subtracting the constant-flux radial backbone, the residual endpoint source is frozen as a bounded reset-decompression cap body plus a finite support-edge closure component. The package remains non-live on baseline and dense ledgers; the dense source regeneration has 90,857 rows, zero positive live packet-norm samples, and max live packet norm about $-6.333723$. The dense support-edge closure remains tiny, with max closure coefficient about 0.000846 in the dedicated dense stability rung. |
| Beta-memory endpoint receiver | The representative effective receiver grammar uses a beta-memory negative-$l$ endpoint receiver. The beta075 $p003\_mid$ receiver is sealed for the engineered $V = 5$ operating point, live-clean in dense promotion, and live-clean in full-stride source-sector checks. The beta100 release-timing stress condition uses the same grammar with service-specific retiming as a stress anchor. |
| Endpoint source class and current regulator | The frozen endpoint source uses a regulated anisotropic heat/current medium, because scalar-only and ordinary Type-I descriptions are algebraically too narrow for the radial heat/current block. A small non-live $\rho + p_l$ regulator removes the flux-dominant radial-block rows with regulator/source burden about 0.037723 on the baseline mesh and 0.039030 on the dense mesh, while preserving zero live regulator rows. |

| Diagnostic layer | Representative result |
|---|---|
| Regulated endpoint medium closure | At a 1.10 current-regulator pad, the regulated medium has no post-regulator Type-IV rows, no superluminal or undefined boost rows, and regulator/source burden about 0.041495 on the baseline mesh and 0.042933 on the dense mesh. The constrained field-closure validation carries the non-live angular-pressure channel as an internal medium response: worst normalized angular L1 error is about 0.002977 on the baseline mesh and 0.009524 on the dense mesh, with zero live, regulator-live, and superluminal rows. |
| Covariant endpoint/source tensor identity | The repaired beta075 endpoint/source components close as a covariant finite-difference tensor on the endpoint audit surface, preserving the regulated radial heat/current block, internal angular response, and live exclusion before support exchange is added. |
| Total endpoint/support exchange closure | The sealed $24 \times 14$ support-stress reference supplies a divergence-form support exchange current paired with the endpoint current. Baseline active and allowed residuals are about 0.244987 and 0.190870 of endpoint L2; dense active and allowed residuals are about 0.451388 and 0.358617. Outside residual fractions stay near $10^{-3}$, live residual fractions stay below 0.004, support-tail fractions are negligible, and angular support/residual exchange is zero. |
| Compact support-stress bracket | The compact $22 \times 13$ support-stress basis preserves aggregate active/allowed closure and localization, while the dense local active residual rises to about 0.581325 of endpoint L2. It is therefore retained as a compactness bracket, with the $24 \times 14$ basis serving as the closure reference. |
| Support-sector robustness gate | The sealed $24 \times 14$ support-stress reference clears the total endpoint/support closure baseline and dense gates with dense local PF residual about 0.544567 against the 0.55 gate. The margin is tight, about 0.009878 fractionally, while the compact $22 \times 13$ bracket misses the promoted dense local gate with local PF residual about 0.597527. Live support fraction remains zero, outside support fraction is about $1.802 \times 10^{-5}$, minimum $h_{\mathrm{reg}}$ is about $4.036 \times 10^{-9}$, and p99 heat-flux ratio is about 0.975555. |
| Reduced principal-symbol gate | The fixed-background endpoint/support principal-symbol diagnostic has real finite in-cone characteristics and complete eigenbases on the promoted baseline and dense $24 \times 14$ audit rows. The dense run has 28,359 active rows, zero hard gate-crossing rows, 56 boundary-near rows, minimum cone margin $7.881 \times 10^{-5}$, p01 cone margin 0.017633, p99 heat ratio 0.975555, and maximum $\psi$ about 5.064348. |
| Principal-symbol sensitivity | Nominal dense boundary rows pass hard local characteristic gates, with heat-current stress marking the tight local boundary. A $+7.5 \times 10^{-5}$ raw heat-current stress still has zero hard gate-crossing rows, while $+10^{-4}$ reaches the first reset-decompression/support-edge hard gate crossing with minimum transport margin about $-2.017 \times 10^{-5}$. Regulator depletion preserves hard gates through one-percent regulator scale and reaches the regulator boundary when driven exactly to zero. |

| **Diagnostic layer** | **Representative result** |
|---|---|
| Bounded rapidity transport pilot | A raw $+10^{-4}$ heat-ratio perturbation crosses one reset-decompression/support-edge local gate. Expressing the same stress in bounded transport rapidity clears the hard gates with zero hard gate-crossing rows, minimum cone margin $2.252 \times 10^{-5}$, minimum transport margin $2.281 \times 10^{-5}$, and p99 heat ratio below one. The larger $+5 \times 10^{-4}$ rapidity stress is over-tight and crosses one row gate, identifying the scale for support-source limiting. |
| Rapidity budget and advection non-concentration | On the 56 dense adversarial rows, every observed $O(10^{-4})$ rapidity source kick fits the row-local budget. The tightest row is reset-decompression/support-edge row 4605, with baseline $\psi = 5.064348$, budget $\Delta\psi = 2.183566$, observed $\Delta\psi = 0.626342$, and observed budget fraction 0.286844. Reduced $1 + 1$ inward and outward advection on the bottleneck slice keeps the observed impulse under budget with limiter inactive; the larger reference kick is classified by the budget guard. |
| Full-package and full-domain support-source evolution | Scoped support-edge source-profile normalization keeps the V5 full-package source-coupled sweep inside the observed-amplitude rapidity budget, with observed outward/inward max budget fractions about 0.289 and 0.373. The sealed full-domain fixed-background evolution has zero live or packet-live evolved rows, zero observed over-budget rows, inactive observed limiter behavior, and maximum observed budget fraction about 0.121201. The deliberate large-amplitude full-domain stress case remains inside budget with maximum budget fraction about 0.604677. |
| Service-coordinate aligned source timing | Service-coordinate source release governs the action-level fixed-background timing class. All tested aligned pulse widths and directions pass, including one-step aligned pulses, with maximum budget fraction 0.742835 and no state/source amplification. The aligned-envelope certificate extends this result to any common nonnegative temporal kernel over service-ordered bins by bounding the kernel response with the one-step service-aligned basis response. |
| Formal source-family equation package | The beta075 source family is written as a regulated anisotropic heat/current medium entrained to a radial director and coupled to a localized support reservoir. The package closes the endpoint tensor identity at about $6.59 \times 10^{-17}$, constrained field closure at about 0.009524, support stroke exchange at about 0.449018/0.5, aggregate total-conservation restoration at about 0.449018/0.5, angular exchange absence at zero, support-tail localization at about $6.36 \times 10^{-6}/0.001$, live support exclusion at zero, and observed discretization robustness at about 0.892161/1. |

| **Diagnostic layer** | **Representative result** |
|---|---|
| Source-family validation | The derivative support-reservoir operator preserves the hard formal equation gates on the baseline and dense V5 fixed-background ledgers. The recomputed dense source-family symbol retains minimum cone margin $7.881 \times 10^{-5}$, p01 cone margin 0.017633, and zero hard symbol gate-crossing rows. The local exchange-shape constants remain explicit: maximum local power-channel shape about 0.609031, maximum local radial-force-channel shape about 0.475062, and dense local support closure about 0.544567/0.55. |
| Cross-surface source-family robustness | The same regulated director/support-reservoir validation preserves hard formal-symbol gates, live-exclusion gates, and total-conservation gates on the available sealed baseline V5 and sealed dense V5 surfaces. The sealed dense V5 surface remains the tight-margin reference surface for cone margin, local P/F shape, and total support closure, while non-V5 service-rating ladders are treated as separate calibration and boundary diagnostics. |
| Fixed-background energy-estimate certificate | The regulated director/support-reservoir source family carries a symmetric energy estimate on the available complete surfaces. The identity symmetrizer is positive definite, the symmetrized principal matrix is symmetric to numerical precision, and the in-cone energy-flux margin remains positive with minimum margin $7.881 \times 10^{-5}$. The largest lower-order work constant is about 2.049051/2.5 on the sealed dense V5 surface, with support work about 0.898931, local exchange about 0.609031, and support closure about 0.544567. |
| Energy-constant audit | The lower-order work constant is classified as a stable limiting theorem constant for the current package. Sealed dense V5 has value about 2.049050946 against the configured hard bound 2.5, leaving absolute headroom about 0.450949054 and utilization about 81.96%. The constant is carried as a thin but positive engineered V5 margin rather than as evidence for redesign at the V5 operating point. |
| Affine-reparameterized null-accumulation accounting | Semilocal radial-null windows are applied to live, handoff, endpoint, and source-sector scopes using an explicitly reparameterized affine window coordinate. The repaired beta075 *rematch_w6_t1p5* package has zero all-window and zero scoreable benchmark-floor violations across 137,064 full-stride windows at affine widths 0.5, 1.0, and 2.0 in both component-sector and intermediate residual-inclusive modes. |
| Finite-domain radial ANEC alignment | Longest-available in-domain affine radial-null trace integrals locate the remaining averaged-null burden in source-completion residuals and plus-branch handoff trim. Residual and live-handoff separation leaves the explicit non-live sector package close to averaged radial-null balance, so this component strengthens the feasibility map while keeping ANEC as an alignment target rather than a certified claim. |

| Diagnostic layer | Representative result |
|---|---|
| Causal-carrier reachability | The shell/throat carrier band contains a measured GZ-like cone signature with measured branch-crossing structure. Entry-side causal floods from boundary and support-edge seed sets record zero service-stage and carry-stage entry-to-live-packet hits in the sampled ledgers. |
| Branch-band radial escape | In the promoted beta075 receiver setting, live packet, packet geometry, main carrier, support plant, and live branch-band seeds reach exterior radial boundaries on both radial branches. Extended $s = 12$ and $s = 15$ confidence domains give radial escape for all 1056 core-mask seed120 traces while the local branch-crossing diagnostic remains present. |
| Dense finite-bundle carrier transport | In the tensor-consistent beta-collar generator setting, the conservative $rematch\_w6\_t1p5$ lead preserves 136/136 dense-ray radial escape, 136/136 recovery from both-shrinking intervals, zero sustained-to-end both-shrinking rays, and 0/8 caustic-like bundles on the $s = 15$ confidence domain. The worst all-both $l$-width ratio is about 0.275792, compared with about 0.009603 in the corresponding unmitigated dense-bundle carrier run. A wider $rematch\_w8\_t2p0$ bracket also gives 0/8 caustic-like bundles and worst all-both $l$-width about 0.394689. |
| Scheduled ADM probe evolution | Prescribed-ADM probes through $\alpha, \beta, \gamma_{ll}, \gamma_{\Omega\Omega}$ preserve radial, off-axis, and small-bundle branch-band escape. Reset-decompression wake-tail traces remain exterior to live service and record zero live re-entry in the confidence run. |
| Service-time advantage ledger | The $s = 15$ exterior-null service-time ledger rates the prescribed schedule as operationally favorable. Prepared service time is about 3.045; the schedule-factor A-to-B proxy gives distance about 7.822491 and advantage ratio about 2.568962, while the packet-coordinate proxy gives distance about 3.755436 and advantage ratio about 1.233312. With request-triggered setup included in the service clock, the ratios remain favorable at about 2.487278 and 1.194097. |
| Beta100 release-timing stress rating | The high-amplitude release family supplies a release-timing stress condition for the same receiver grammar. The representative V5 operating embodiment remains rated on V5 packet safety, source localization, compact handoff behavior, and semilocal finite-window null margin. |

The combined evidence defines the design interpretation. The small support-shell overlay is a reliable reduced-harness support-routing component. The larger 4D source-relief objective uses coupled metric timing, radial gradient matching, derivative-smooth release choreography, compact pressure-aware handoff shaping, semilocal null-accumulation accounting, packet-local radial support placement, and explicit source-family assignment. Clock-lapse timing is the strongest default partner for aggregate burden reduction, raised-cosine annular support is an important matched-strength comparator, and same-window throat-capacity is reserved for alternate intrinsic-window accounting. The representative V5 end-to-end operating embodiment is the blended trailing-edge release-choreography family with finite-bundle beta-collar widening, two-feature packet-local radial support, entry service gating, compact entry/catch/edge handoff support, dense source-sector closure, sealed beta075 $p003\_mid$ endpoint receiver accounting, affine-reparameterized radial-null margin, causal-carrier gates, total endpoint/support exchange closure, reduced support-sector characteristic checks, service-coordinate source-release timing, and exterior-null service-time rating. It is packet-safe under smoothing and refinement, preserves dense branch-band bundle transport in the carrier-collar audit, records favorable A-to-B service-time advantage, and localizes demanded-source burden into separately diagnosed live handoff trim, non-live radial string-cloud support, coupled endpoint/junction structure, angular capacity, current relaxation, support stroke/stress completion, reduced bounded-rapidity transport response, and reset-domain wake-tail restoration. The sealed beta075 endpoint/support realization is a finite non-live source package with a regulated anisotropic heat/current medium and a phase-aware support stroke/stress-potential sector that pass the finite-difference closure gates; as a reduced fixed-background support-sector system it also passes local principal-symbol, bounded-rapidity, row-budget, observed-amplitude advection, full-package source-coupling, full-domain evolution, service-coordinate aligned-envelope timing, formal source-family equation validation, engineered V5 surface robustness, and fixed-background energy-estimate checks. The present claim boundary covers finite-difference endpoint/support closure plus reduced support-sector transport, aligned source-timing, fixed-background regulated director/support-reservoir source-family validation gates, and symmetric energy-estimate certification on the complete engineered V5 surfaces. The service-rating ladders at $V = 2.5$ and $V = 10$ sit outside that claim boundary and identify the next $V$-aware source-law obligations.

## 14. SOURCE-FAMILY TARGET ARCHITECTURE

The source-family target architecture translates the demanded-source ledger into source roles and endpoint/support closure tests. It retains the prescribed metric as the geometric service program and assigns demanded stress into a compact set of source targets. Non-minimally coupled scalar and traversable-wormhole source constructions provide useful admissibility references for energy-condition violation, averaged null energy, and trans-Planckian source thresholds [6]. The target grammar is

$$\mathcal{S}_{\text{target}} = S_0 + J + R + C_{\text{live}} + G + D_H, \tag{61}$$

where $S_0$ is the constant-flux radial string-cloud backbone, $J$ is the coupled endpoint/junction layer, $R$ is small core-body residual leakage, $C_{\text{live}}$ is live handoff trim, $G$ is angular-capacity support, and $D_H$ is non-live current relaxation. This grammar assigns infrastructure, handoff, and endpoint responsibilities. Physical components can share several roles and overlap spatially. The support-reservoir target receives the fitted exchange paired with the endpoint current; its independent tensor belongs to the joint construction.

| **Target sector** | **Assigned source role** | **Representative diagnostic** |
|---|---|---|
| $S_0$ radial string-cloud backbone | Non-live radial-tension scaffold for the dominant A/B/I support infrastructure | $\rho > 0$, $p_l \approx -\rho$, constant areal flux |
| $J$ endpoint/junction layer | Coupled support-edge shoulder and reset/decompression cap carrying radial trim, angular capacity, and current relaxation | selected-null deficit, current, $p_\Omega$, finite endpoint/support closure |
| $R$ core-body residual leakage | Small residual core-body source after subtracting the radial backbone | residual selected-null and pressure leakage |

| | | |
|---|---|---|
| $C_{\mathrm{live}}$ live handoff trim | Packet-facing handoff trim for angular/current, radial-null, and radial-pressure service rows | live fractions, packet norm, handoff localization |
| $G$ angular-capacity support | Non-live angular and throat-capacity support outside the packet corridor | $p_\Omega$ support and shell/throat placement |
| $D_H$ current relaxation | Non-live reset/background and distributed current relaxation | $j_l$, transfer accounting, exchange closure |

The dominant non-live radial support has the local algebra of a radial-tension scaffold. A representative unit-frame fit gives

$$\frac{p_l}{\rho} \approx -0.999288, \qquad \frac{j_l}{\rho} \approx -0.000485, \qquad \frac{p_\Omega}{\rho} \approx 0.001078, \tag{62}$$

with zero live scaffold fraction in the source-target ledger. The ideal density and areal flux are

$$\rho = \mu = \frac{\Phi}{\gamma_{\Omega\Omega}}, \qquad \Phi = \rho\gamma_{\Omega\Omega}. \tag{63}$$

The nearly constant $\Phi$ gives the backbone a concrete string-cloud realization target. In the representative dense source-family target ledger, A/B/I carries about 94.03% of that ledger's burden; the selected-null deficit after scaffold subtraction is about 1.338% of the scaffold, with zero live fraction and concentration at non-live support-edge and reset/decompression structures. The areal flux $\Phi$ has weighted relative mean deviation about 0.0904%. These are target-ledger measures; actual supplied load shares use the independently normalized physical component tensors.

The endpoint/junction layer is the primary coupled realization target after the radial backbone. It combines the support-edge shoulder and reset/decompression cap into one source role because the same endpoint region carries residual radial trim, angular endpoint support, and current relaxation. In representative junction accounting, the coupled $J$ layer carries selected-null deficit about 1.010455, current about 0.381351, and angular support about 3.961788, with zero live rows and zero live selected-null burden.

The archived beta075 V5 endpoint fit uses a bounded no-tail reset-decompression cap and a finite-width support-edge component. On the dense $377 \times 241$ ledger it retains its non-live placement and bounded coefficients. The fitted tensor and geometric $J$ are distinct quantities: their replacement difference is a source target carried explicitly in the joint assembly.

The fitted endpoint tensor has an anisotropic heat/current representation. A small non-live regulator changes the $\rho+p_l$ pair while retaining $\delta j_l = 0$ and the angular and placement constraints. At the selected 1.10 pad, the recorded regulator/source burden is 0.041495 on the baseline mesh and 0.042933 on the dense mesh, with zero live regulator rows. The regulator is included in the fitted medium tensor. Its physical rest-frame energy and admissibility margins use the corrected signed-enthalpy eigensystem.

The same fitted medium carries the non-live angular-pressure channel as an internal response. Holding the cap, support-edge component, and regulator fixed, its recorded worst normalized angular L1 errors are 0.002977 and 0.009524 on baseline and dense meshes. The covariant tensor-identity check verifies reconstruction of that fitted medium. The archived boost-dependent checks require corrected reevaluation; the geometric source and fitted tensor remain independently reproducible.

The support stroke/stress-potential basis fits the complementary endpoint exchange current. Its $24 \times 14$ reference passes the recorded baseline and dense aggregate, local, localization, live-exclusion, support-tail, and angular-exchange residual gates. The $22 \times 13$ basis remains a compactness bracket because the dense local residual selects the larger reference. These measurements constrain a reservoir's required exchange and effective response. Evaluating its independent stress tensor supplies the absolute energy, pressure, and algebraic class that the divergence fit leaves open.

The formal source-family package represents the endpoint and its exchange through a regulated medium, radial director, and support reservoir. The fitted endpoint tensor has the anisotropic heat/current form

$$T_{\mu\nu} = \rho u_\mu u_\nu + p_l s_\mu s_\nu + p_\Omega Q_{\mu\nu} + u_\mu q_\nu + u_\nu q_\mu, \tag{64}$$

with $u^\mu$ the chosen medium frame, $s^\mu$ its radial director, $q_\mu = j_l s_\mu$, and $Q_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu - s_\mu s_\nu$. On resolved non-diagonal Type I rows, the effective heat ratio satisfies $2j_l/|\rho + p_l| = \tanh\psi$. This ratio supplies the reduced operator's transport coordinate; its relation to the energy-frame boost is given in Section 15. The support-source response uses the phase-local law $S_\psi = \kappa_{\text{phase}} B_{\text{service}} R_\psi$. The target exchange equations are

$$\nabla_\mu T^{\mu\nu}_{\text{endpoint}} + J^\nu_{\text{support}} = 0, \qquad J^\nu_{\text{support}} = P u^\nu + F s^\nu + J^\nu_\perp, \qquad \nabla_\mu T^{\mu\nu}_{\text{support}} = J^\nu_{\text{support}}, \tag{65}$$

with the audited angular support exchange absent on the accepted surface. The support reservoir has a derivative storage/stress form with $\partial_\tau \Phi_{\text{support}} = \Pi_{\text{support}}$, spatial support derivatives, damping, and lower-order $P, F$ exchange drives.

The fixed-background source-family validation preserves this equation package on the baseline and dense V5 surfaces. The recomputed source-family symbol retains real complete in-cone characteristics, zero hard symbol gate-crossing rows, dense minimum cone margin $7.881 \times 10^{-5}$, and dense p01 cone margin 0.017633. The hard package gates include endpoint tensor identity, constrained field closure, support stroke exchange, aggregate total-conservation restoration, angular-exchange absence, support-tail localization, live support exclusion, and observed discretization robustness. The source-family margins are explicit design constants: minimum phase-local scale 0.092617, local power-channel shape 0.609031, local radial-force-channel shape 0.475062, and dense local support closure 0.544567/0.55.

Cross-surface source-family robustness evaluates the same regulated director/support-reservoir package on the available complete engineered source-family surfaces: sealed baseline V5 and sealed dense V5. Both surfaces preserve the hard formal-symbol, live-support exclusion, and total-conservation gates. The sealed dense V5 surface remains the tight-margin reference surface, with the same cone-margin, local exchange-shape, and local support-closure constants carried into theorem-style packaging. The $V = 2.5$ and $V = 10$ ladders are not promoted to proof surfaces; they identify where a future service-aware source law must retune medium/support closure and high-service causal margin.

The fixed-background energy certificate equips the regulated director/support-reservoir equations with a symmetric energy estimate on the available complete surfaces. With

$$E[U] = \frac{1}{2}\int U^T H U \, dV, \qquad H = I \tag{66}$$

on $U = (h, \psi, \chi_\Omega, \pi_\Omega, \Phi_{\text{support}}, \Pi_{\text{support}}, n_l)$, the symmetrized principal block is symmetric to numerical precision and the energy flux remains inside the service cone. The certificate carries the support exchange channels $P, F$, total support closure, and scheduled phase-local source response as explicit lower-order work constants. The sealed dense V5 surface sets the current sharp constants: minimum flux margin $7.881 \times 10^{-5}$, lower-order work constant about 2.049051/2.5, support work about 0.898931, local exchange about 0.609031, and support closure about 0.544567/0.55.

The energy-constant audit classifies the recurring work constant as a stable limiting theorem constant of the present fixed-background package. On sealed dense V5, the value 2.049050946 leaves absolute headroom about 0.450949054 against the hard bound 2.5, or about 18.04% of the hard bound. The decomposition is distributed across support work, local $P/F$ exchange shape, and support closure, with respective shares about 43.87%, 29.72%, and 26.58% of the worst work constant. This places the reset-decompression endpoint-junction and support-edge rows as the theorem-margin focus for the first-order coupled consistency rung.

The source-construction boundary therefore comprises a reproducible demanded tensor, a fitted endpoint tensor, a complementary exchange-current fit, and effective fixed-background transport and energy estimates. The characteristic and work constants above belong to those specified operators. A complete physical assembly supplies the remaining component tensors, including the reservoir, retains all replacement residuals, and recomputes causal-frame and coupled-response gates. The bounded semi-classical material/quantum investigation further supplies a regular material candidate and an absolute quantum-stress comparison on its registered smooth static geometry. Its insufficient supplied source constrains that realization while preserving the broader component responsibilities.

The beta-memory endpoint receiver supplies the effective receiver mechanism within this grammar. Its amplitude is tied to accumulated beta-release transfer and applied through a negative-$l$ support-edge receiver channel. The representative beta075 $p003\_mid$ receiver provides the sealed effective receiver setting for the V5 source-target architecture: it is live-clean, endpoint-effective, and positive-margin in the full-stride affine-reparameterized radial-null package. The beta100 release-timing stress condition uses the same grammar with service-specific receiver retiming; the post-release 1.5 anchor gives positive selected, current, and angular transfer accounting while preserving the live gate and SNEC companion margin.

## 15. COUPLED SOURCE CONSTRUCTION AND JOINT COORDINATION

The source construction retains four distinct descriptions: metric controls specify service targets; the role ledger assigns demanded channels by location and phase; physical components supply stress and interaction; and state variables determine their response. A physical component can serve several roles, while several components can share one load. The component cross-reference and coordination report connects this architecture to the tested source families and their registered backgrounds.

### 15.1 Responsibilities and shared loads

| **Role** | **Responsibility in a physical assembly** |
|---|---|
| Standing substrate and $S_0$ | Pre-existing energy, dominant radial tension, director orientation, and continuous load paths into the surrounding plant. |
| Angular jacket and $G$ | Transverse support, angular restoring response, and the stress combination that shapes clock curvature. |
| Endpoint $J$ and $D_H$ | Radial trim, angular response, current transport and relaxation, and transfer into stored support state. |
| Live handoff $C_{\text{live}}$ | Separate angular/current, radial-null, and radial-pressure responses in entry, catch, rematch, and release, with hard infrastructure excluded from the live corridor. |
| Reservoir and receiver | Counted energy, momentum, strain, release memory, power/radial-force exchange, and reset readiness. |
| Finite transition and quantum environment | Smooth mechanical and optical matching, confinement/screening, absolute quantum stress in both null directions, and reciprocal material force. |
| Service controller | Admissible forcing, shared timing, and readiness constraints, with its work accounted through the plant or exterior flux. |

The engineering analogs in the component cards describe stiffness, routing, impedance, storage, and sensing responses. A gravitational source candidate supplies its actual stress, scale, and couplings. In particular, an architected lattice's load-path analogy leaves its attainable energy-to-tension ratio as a material requirement.

The distinction between bulk support and opening is exact in the static proper-distance chart

$$ds^2 = -A(l)^2dt^2 + dl^2 + R(l)^2d\Omega^2.$$

Here $R(l)$ is areal radius, distinct from the residual role $R$, and primes are proper-distance derivatives. In this subsection $\rho, p_r, p_t$ denote geometrized stresses $G_{\mathrm{phys}}$ times the corresponding physical stresses, with $c = 1$. At a strict smooth throat,

$$p_r\big|_0 = -\frac{1}{8\pi R_0^2}, \qquad (\rho + p_r)\big|_0 = -\frac{R_0''}{4\pi R_0} < 0. \tag{67}$$

The first expression sets bulk radial tension; the second sets signed opening support. An ideal contribution $I(1,-1,0)$ changes the density/tension budget while leaving $\rho + p_r$ unchanged. Separate static conservation requires $IR^2$ constant; termination or varying flux transfers force through angular stress or component exchange.

Clock shaping and total force balance impose

$$(R^2A')' = 4\pi R^2A(\rho + p_r + 2p_t), \tag{68}$$

$$p_r' + \frac{A'}{A}(\rho + p_r) + 2\frac{R'}{R}(p_r - p_t) = 0. \tag{69}$$

Consequently radial load, clock profile, and transition stresses must be selected together. The audited smooth static seed has negative radial-null balance

$$B_- = 4\pi \int \frac{R}{A}\max[-(\rho + p_r), 0]\, dl.$$

Over the rail-coordinate domain $-40 \le x \le 40$, 96.69% lies outside $|x| \le 2$, and some outer points also require negative angular null stress. These measured bins belong to that seed, which includes a condensate exterior and proper-distance smoothing. The positive-side inherited tail itself carries source; both exterior completions therefore enter the construction.

The source-role audit gives the relaxed condensate only 0.612% of the demanded throat tension. Its assigned quantum remainder carries 99.388%, including most of the bulk support. Restoring an explicit backbone addresses that allocation. Independently, the neutral scalar's tested finite-regulator controls are short by approximately 186,000–204,000 in the optimistic integrated opening comparison, and its signed contribution opposes opening overall on that seed. Ordinary components with nonnegative null stress preserve this bound at fixed geometry and quantum profile. Continuum extrapolation and a full joint fixed point remain unresolved. A physical assembly therefore screens actual bulk shares and both residual null directions under its common normalization.

The counted support comparison uses an explicit backbone with tensor $\Phi(l)R^{-2}(1,-1,0)$. It carries 95% of throat tension alongside the saved condensate, independent radial/angular ideal Casimir targets, and a fitted host obeying the dominant energy condition. Its radial quantum weight at the throat is $3.41474 \times 10^{-7}$. The backbone's finite termination requires radial exchange $-\Phi'/R^2$. Across the retained interval, angular response and termination retain larger loads: the optimized quantum proper-energy magnitude grows from 23.38925 to 27.78549 as the backbone allocation grows from zero to 95%. Requiring only nonnegative host null stresses gives the weaker magnitude bound 2.88833 in the latter case. The difference measures the additional energy cost of angular pressure under the specified host pressure bound. These quantities define a counted target allocation; its material action, absolute quantum supply, and coupled response remain construction inputs.

Ideal longitudinal conformal channels provide a separate integrated source gate. Write $a = \log A$, $k_Q = \eta\mathfrak{c}/(12\pi)$ for total central charge $\mathfrak{c}$, $C = 4\pi^2/L_{\mathrm{opt}}^2$, and $W = \exp[-(R^2 - R_0^2)/(2k_Q)]$. Their radial quantum null stress is $k_Q(a'' - C/A^2)/(4\pi R^2)$. The remaining aggregate obeys the identity

$$C\int \frac{W}{A^2}\, dl - \int \frac{WRR''}{k_Q}\, dl - [Wa']_-^+ = \frac{4\pi}{k_Q}\int WR^2H_{\mathrm{rem}}\, dl. \tag{70}$$

For $H_{\text{rem}} \geq 0$, the right side is nonnegative. This aggregate can include independent bulk, angular, and host sectors, including an ideal angular Casimir pair with zero radial null stress. At fixed $R(l)$ and endpoint clock slopes, interior clock reshaping preserves the two demand terms. For $k_Q/R_0^2 = 0.01$ and channels spanning $-7 \leq x \leq 7$, an optimistic bound over all clock shapes between 50% and 150% of the saved lapse supplies at most 4.17% of the required balance, even with zero exterior optical return length. This exploratory clock range leaves operational admissibility to the service gates. The exclusion concerns the stated channel law and retained radius profile; a changed radial transition or an additional negative radial-null source changes the construction problem.

The archived-control comparison applies this gate to thirteen parameter transfers on the repaired beta075 static slice at phase 0.745, retaining gravitational normalization and total central charge across candidates. The controls include support radius, angular width, radial core/ring/skirt shape, and receiver/beta comparisons. At $k_Q/R_{0,\text{reference}}^2 = 0.01$ and $-7 \leq x \leq 7$, the strongest tested improvement increases the original-clock supply ratio from 0.00461481 to 0.00462933 by increasing radial core gain from 0.05 to 0.06. Its optimistic 50% clock-box ceiling is 0.0416640. The same necessary gate retains every reference exclusion across the registered strength and interval ladder. Receiver memory is zero on this construction phase; the beta-collar control changes carrying flow while preserving the static lapse and spatial metric.

Increasing support radius from 1.75 to 2.05 lowers local negative-density and radial-null peaks while increasing the proper-volume negative-density magnitude by 15.8% and absolute angular-pressure integral by 27.4% over $-7 \leq x \leq 7$. At the fixed coefficient 0.01, its opening ratio falls by 8.45%. All thirteen transfers retain negative packet norm in the scoped scalar service screen. Stronger coefficients admit some short-interval necessary balances, including the reference geometry, while the longer original-clock intervals remain excluded on the tested ladder. These results bound the usefulness of the evaluated controls for this longitudinal source law; full source assembly retains its independent tensor, quantum-state, and exchange requirements.

The material-confined fermion evaluation uses a real canonical scalar with $m = y\chi$, vacuum amplitude $v = 12/6.8$, and proper interface width $d = 1$ to enclose the retained static rail at areal radius 6.8. With the gravitational normalization fixed, its resolved positive-frequency bound catalogs contain 74, 302, and 1,222 angular/radial multiplets for $y = 1, 2, 4$. Every occupied multiplet has opposing integrated opening $B_F = -4\pi \int R(\rho_F + p_{r,F})\, dl/A$. Independent occupations therefore supply at most zero opening; filling each complete catalog gives $B_F = -0.00449690, -0.06977054, -1.10171728$, respectively. Over $-96 \leq x \leq 96$, the geometric requirement is 1.99966778 and the canonical confining scalar adds $-0.00428149$. Boundary extension preserves each catalog and opening sign. These tensors measure stationary occupied-state increments on the specified scalar profile; absolute vacuum stress, curved material equilibrium, and active evolution remain independent construction requirements.

The finite curved cavity evaluation uses two disjoint canonical scalar layers with quantum mass potential $g(\chi_1^2+\chi_2^2)$. Their ultraviolet-finite interaction vacuum supplies helpful opening at all nine selected placements on the repaired native static rail. The counted mirror gradients exceed that interaction at portal couplings $g = 1, 4, 10$, with inferred curved crossing couplings between approximately 17,438 and 24,191. Keeping the physical mirror height and thickness fixed while reducing the gap sixteenfold increases the integrated interaction support by approximately 1.67. Preserving optical similarity instead raises the mirror gradients alongside the quantum interaction. These comparisons grant zero additional opening cost to the holding sector. The isolated-layer absolute vacua and material equilibrium remain separate contributions and construction requirements.

The short magnetic circuit evaluation grants $c = qN_f$ ideal longitudinal channels to flux $2\pi q$ in a complete local loop on the same native rail. With gauge action $-F^2/(4e^2)$ and radial tangent $t_r$, its Maxwell contribution satisfies $\rho_B + p_{r,B} = B^2(1 - t_r^2)/e^2$. Thus radial segments saturate the null projection while bends add positive load. Including the longitudinal Casimir term and Weyl anomaly gives helpful quantum opening on 303 of 440 paths. The quantum-plus-magnetic opening remains negative in all 23,760 source combinations for each of two field prescriptions: constant magnitude and minimum allowed local magnitude with field spreading. The smallest inferred crossing has $N_f e^2/(16\pi^2) = 17.971$, compared with the preferred perturbative limit 0.1. This leading-channel comparison grants zero additional radial-null cost to currents and material supports; the full transverse spectrum and material equilibrium remain further conditions. The source comparison retains the separate backbone, angular, transport and reservoir roles. A further magnetic adaptation requires a specified global return geometry or a different relation between channel count and confining stress; published global magnetic constructions retain their distinct topology and clock profiles.

### 15.2 Tensor, state, and exchange accounting

The tensor-assembly pre-flight distinguishes role coverage from source supply. Channel-assignment rows repeat the local demanded tensor; independent component stresses require an explicit partition. The stored $S_0/J/R$ partition reconstructs its declared subset, approximately 34% of the full pre-flight grid. The completed endpoint medium differs from geometric $J$ by 49.62% and 47.98% in the stated J-normalized channel norm, or 2.99% and 2.87% of the full geometric-source norm. That difference, $\Delta T_J$, remains visible alongside core residual $R$. The current regulator is already included in the endpoint medium.

The support audit supplies a complementary fitted exchange current. A reservoir tensor must additionally supply its energy and stresses: a divergence-free addition leaves the exchange fit unchanged. With every interaction term and finite counterterm assigned once in a common physical normalization, the construction requires

$$T^{ab}_{\text{total}} = \sum_i T^{ab}_i, \qquad \nabla_a T^{ab}_i = F^b_i, \qquad \sum_i F^b_i = 0, \tag{71}$$

$$\mathcal{R}^{ab}_G = \frac{G^{ab}}{8\pi G_{\text{phys}}} - T^{ab}_{\text{total}}. \tag{72}$$

The sum includes quantum expectation values and the chosen interaction allocation. A fixed convention can place local gravitational counterterms on the left side. The residual $\mathcal{R}^{ab}_G$ measures source still to be supplied. Boundary fluxes and control work have explicit records; a local tensor balance alone supplies no general global-energy conservation law.

The implemented $1 + 1$ transport advances a rapidity increment. The reduced $3 + 1$ extension uses constraint-driver proxies, while the moderate angular/time response model advances scheduled Fourier-weighted amplitudes with bounded feedback. These supply effective response controls for a later signed-stress and metric evolution.

The effective state vector

$$(h, \psi, \chi_\Omega, \pi_\Omega, \Phi_{\text{support}}, \Pi_{\text{support}}, n_l)$$

separates enthalpy, heat ratio, angular response, storage, and director variables. Its $h$ differs from the condensate Higgs amplitude. For a resolved non-diagonal Type I radial block, write $H = \rho + p_r$ and $q = 2j_r/|H| = \tanh\psi$. In the implemented current convention the energy-frame boost is

$$v_L = \frac{2j_r}{H + \operatorname{sgn}(H)\sqrt{H^2 - 4j_r^2}}, \qquad \eta_L = \tfrac{1}{2}\operatorname{sgn}(H)\psi. \tag{73}$$

Thus the bounded heat ratio and physical boost have distinct meanings. At $j_r = 0$ the diagonal tensor is Type I, including $H = 0$ degeneracy; the ADM frame supplies a valid energy frame without evaluating 0/0. The corrected causal classifier supplies the rest-frame-dependent checks. Its Type IV findings obstruct the intended ordinary rest-frame reconstruction where that class is required; quantum-inclusive source models specify their own admissibility conditions.

The regularized metric tests retain active Type IV demand near static enthalpy zeros. Current and diagonal stress must therefore evolve together where an ordinary material rest frame is required. The permitted source class and its causal eigensystem form an explicit joint constraint.

### 15.3 Reciprocal dynamics and construction order

The tested static-start prescriptions require earlier material response or revised initial and boundary data. On those reset paths, angular demand begins at order $u^{n-2}$ while the initially empty moving material responds at order $u^{2n-2}$. Stored initial state and its constitutive response provide one construction direction; changed initial motion, geometry evolution, or boundary supply provide other degrees of freedom. Independent particle and entropy currents give a tested local transport basis, and the action framework of Comer, Andersson, Celora, and Hawke supports state-dependent dissipative constitutive choices [8]. Their specialization must satisfy its entropy-production condition and share exchange with a counted support reservoir.

Mechanical and optical response also share material fields. In the screened condensate, the Higgs amplitude sets potential energy, gradient load, and the neutral scalar's mass. The registered portal interaction obeys

$$\nabla_a \langle T_Q^{ab} \rangle_{\mathrm{ren}} = -\frac{1}{2} \langle \chi^2 \rangle_{\mathrm{ren}} \nabla^b V_\chi. \tag{74}$$

The material equation receives the opposite force with the same renormalization conditions. A second field is useful when its coupling supplies a needed independent response, with its own stress included. Assigning the condensate's electric, potential, or gradient terms to named roles partitions its existing tensor. Combining a microscopic tensor and a hydrodynamic description of the same matter requires an explicit matching prescription.

A stationary quantum ground state defines a standing reference. Active service requires state evolution or a justified causal response approximation that includes excitation and work. The closed-time-path framework supplies real causal expectation-value equations for quantum backreaction [9]. Smooth internal cuts within one material model share field values and normal derivatives; general material interfaces instead match the appropriate canonical fluxes and tractions, with surface actions counted when present.

The joint construction follows five dependent stages:

1. Specify fields, charges/fluxes, interactions, quantum state, renormalization, initial storage, both exteriors, and allowed metric freedom. Assign bulk, angular, current, and interface duties before evaluating their supplied tensors.
2. Screen the full assembly for actual load shares, both null directions, interface costs, and necessary integral balances. Quantum modes include their complete optical paths and material response. A failed necessary sign or scale condition ends that specified candidate.
3. Solve the standing geometry and material boundaries together while recomputing quantum stress and reciprocal force. Require simultaneous field, exchange, and Einstein residual control on the same configuration.
4. Evaluate coupled radial/angular material, current, reservoir, quantum, and metric response. Equilibrium pressure balance and the archived fixed-background operator controls provide separate checks.
5. Evolve a bounded startup/hold/release/reset segment. The rail schedule supplies service constraints; source-derived evolution determines the attainable trajectory. Evaluate causal classification, packet/carrier gates, charge and entropy balances, exterior fluxes, and the reservoir's final readiness together.

## 16. CAUSAL CARRIER AND SHELL-THROAT CORRESPONDENCE DIAGNOSTIC

Garattini and Zatrimaylov describe a wormhole–warp-drive correspondence in which a Morris–Thorne wormhole metric can be written in a warp-drive-like form by allowing nonzero intrinsic spatial curvature in the warp-drive metric [5]. Their correspondence identifies a shell/throat layer where an interpolation profile, a shift-like sector, intrinsic metric curvature, and intrinsic metric derivatives enter the same curvature expressions. In the active-rail notation, the support-shell timing window $W_{\rm sh}$ supplies the localized shell/interpolation profile, $\beta$ is the carrying-flow or shift-like sector, and $G, Q$ are the intrinsic support/throat metric components. The active-rail carrier band therefore lies in the nearby design space where wormhole/throat structure and warp-shift structure meet: local radial cones can tilt far enough that one coordinate null branch crosses zero while the stationary-vector indicator changes sign in the same active support region.

Clark, Hiscock, and Larson studied null geodesics in the Alcubierre warp-drive spacetime and found horizon-like conical access structures for superluminal warp-bubble operation [1]. Later geodesic studies of the Alcubierre metric likewise treat null and timelike geodesic behavior as a central optical diagnostic for warp-shift spacetimes [2]. The active rail is a throat-loaded service-carrier architecture, and the CHL comparison supplies a nearby optical diagnostic reference. The comparison is CHL-adjacent: it supplies the relevant diagnostic lesson that shift-like service geometry is evaluated most cleanly at both the pointwise/single-ray scale and the finite-bundle scale. GZ is the closer reference for throat/branch correspondence; CHL is the closer reference for null-geodesic access, cone formation, and finite-bundle optical behavior. The active-rail carrier audit therefore evaluates both the local branch structure and the dense bundle transport through that branch structure.

The superluminal timing comparison supplies a separate chronology-governance requirement. Everett–Roman tube networks and Shoshany–Snodgrass two-warp constructions show that superluminal service links acquire closed-timelike-curve relevance when multiple service legs are composed across incompatible endpoint orderings [3, 4]. The active rail treats that literature as a safety-engineering constraint on service sequencing: the transport channel is armed by a rail-time governor, reset and endpoint-recovery windows are scheduled as service-readiness intervals, and connected rail networks require one shared chronology policy.

A single-channel overlay

$$\beta = \beta_0 + a_\beta W_{\rm sh} \tag{75}$$

therefore supplies a shell-like carrying-flow feature while assigning intrinsic metric response to the $G$ and $Q$ partners. The demanded-source ledger detects burden where $W_{\rm sh}$ overlaps gradients in $G$ and $Q$. The raised-cosine annular bearing shell, delayed support timing, clock-lapse coupling, and radial-support placement treat this overlap as a coordinated carrier layer with timing and gradient matching across shell/throat structure.

The causal carrier diagnostic evaluates the radial null speeds

$$v_\pm(l,\sigma) = -\beta(l,\sigma) \pm \frac{\alpha(l,\sigma)}{\sqrt{\gamma_{ll}(l,\sigma)}}, \tag{76}$$

the stationary-vector indicator

$$g_{\sigma\sigma} = -\alpha^2 + \gamma_{ll}\beta^2, \tag{77}$$

branch-margin maps, shell/throat-overlap scores, entry reachability, radial escape, prescribed-ADM probe evolution, dense finite-bundle carrier transport, and exterior-null service-time rating. This diagnostic measures the active rail as a controlled causal carrier: local one-way branch structure can be present in the carrier band, while the live packet and main carrier are accepted by service-stage reachability, radial escape, finite-bundle transport, and operational A-to-B timing gates.

In the representative beta075 receiver setting, the local branch-crossing structure is measured directly alongside packet norm and source-channel ledgers. The refined beta075 branch band has a sampled live-packet minimum branch margin of approximately 0.002402, with live-packet branch-crossing edges and nonnegative $g_{\sigma\sigma}$ points in the active shell/throat overlap band. This preserves the measured GZ-like cone signature as a real carrier-layer feature and assigns its service consequence to causal reachability and escape gates.

The entry-to-packet reachability gate uses sampled radial causal floods from domain boundaries, support outer edges, and main-support outer edges. These ledgers record zero service-stage and carry-stage entry-to-live-packet hits across the promoted beta075 receiver, beta100 stress cases, receiver-retimed cases, and causal-margin-guard variants. Post-release main-support-edge hits are assigned to reset/wake restoration accounting, where they are tracked separately from live service.

The second paired gate is live-service radial escape. For the promoted beta075 $p003\_mid$ mechanism on the refined long-tail domain, live packet, packet geometry, main carrier, support plant, and live branch-band seeds reach exterior radial boundaries in both radial branches. Increased targeted seed density preserves that result for the live packet, main carrier, support plant, and branch-band families; the branch-band minus families pass through the branch-sensitive region and reach exterior radial boundaries. On the extended $s = 12$ and $s = 15$ confidence domains, all 1056 core-mask seed120 traces reach radial boundaries while the local branch-crossing diagnostic remains present.

The prescribed-ADM probe audit provides the richer scheduled-geometry read. Evolving probes through $\alpha(\sigma, l)$, $\beta(\sigma, l)$, $\gamma_{ll}(\sigma, l)$, and $\gamma_{\Omega\Omega}(\sigma, l)$ preserves radial, off-axis, and congruence branch-band escape. The finite-bundle carrier audit supplies the stronger acceptance criterion: the branch-band preserves neighboring-ray width and ordering at the carrier scale in addition to individual ray recovery. With the tensor-consistent widened trailing-edge beta-rematch collar, the representative $s = 15$ dense congruence audit gives 136/136 dense-ray radial escape, 136/136 recovery from both-shrinking intervals, zero sustained-to-end both-shrinking rays, and 0/8 caustic-like bundles. The conservative $rematch\_w6\_t1p5$ setting gives worst all-both $l$-width ratio about 0.275792, and the wider $rematch\_w8\_t2p0$ bracket gives about 0.394689. The exterior-null service-time ledger supplies the operational A-to-B rating: the representative $s = 15$ schedule gives prepared-service advantage ratios about 2.568962 under the schedule-factor proxy and about 1.233312 under the packet-coordinate proxy, with favorable request-triggered accounting as well. The explicit reset-decompression wake-tail monitor clears monotonically with forward-domain extension, remains exterior to live service, and records zero live re-entry in the tested confidence run. The remaining wake-tail traces belong to reset-domain tail-closure monitoring.

The resulting acceptance language is componentized. The active rail includes a measured GZ-like cone signature as a carrier-band feature, CHL-adjacent branch-band optics as a finite-bundle diagnostic comparison class, live-service radial escape as a prescribed-metric operating property, mitigated finite-bundle carrier transport as a branch-band acceptance gate, exterior-null service-time advantage as an operational rating, and endpoint/wake tail closure as a reset-domain diagnostic. These diagnostics support a design in which the protected live packet corridor is carried by support infrastructure while causal access, source placement, branch-band transport, endpoint timing, and endpoint restoration are evaluated by separate gates. The disclosure therefore states the current prescribed-metric result positively: the repaired beta-collar carrier passes the tested CHL-adjacent branch-band optics gate with zero sustained trapping traces and zero dense caustic-like bundle-collapse flags. Higher-tier causal-structure and source-family realization diagnostics separately address global-horizon exclusion and physical matter-model scope.

## 17. DIAGNOSTIC ARCHITECTURE AND ACCEPTANCE PATH

### 17.1 Reusable Diagnostic Layers

The diagnostic architecture includes the following reusable layers:

1. Exact reduced-field builder and ADM ledger for $\alpha$, $\beta$, $G$, and $Q$, including packet norm, region labels, service labels, $\rho_{\mathrm{ADM}}$, and $j_l$.
2. Standing substrate subtraction using carrying-flow-off or ready-state fields to isolate service-induced $\Delta\rho$ and $\Delta j_l$.
3. V5 support-shell ansatz accounting, selecting the direct annular support-shell window over momentum-following controls.
4. Reduced diagnostic ladder for the small support-shell target, including target gates, rich objective scoring, sign comparison, packet-safety overlay, balance bookkeeping, and ramp classification.
5. Historical V10 release-timing stress checks for the small support-shell target, retained as reduced-target context rather than as a current operating embodiment.
6. Continuous metric-side source-ledger characterization of the support-shell carrying-flow overlay.
7. Coupled timing sweeps adding clock-lapse and rail-stretch partners to the continuous support-shell metric expression.
8. Same-window throat-capacity accounting, identifying $\kappa_Q/a_\beta = 0$ as the default same-window throat-capacity setting.
9. Support-shell shape selection, identifying the raised-cosine annulus as the preferred radial bearing shape at matched peak support-shell strength.
10. Matched-strength robustness ladder for the selected raised-cosine annulus support-shell family.
11. Arbitrary service-factor input generation for reduced ADM ladder hardening and interpolation checks.
12. Packet carve/lapse and annular shoulder sweeps, showing that packet safety and live high-burden channel exposure depend on both causal-margin restoration and support-edge shaping.
13. Packet beta-rematch and release-choreography accounting, including matched hold, minimum-jerk beta fade, coordinated lapse/carve support, and shaped packet-edge beta rematch.
14. V5 floor-union refinement accounting for the release family, showing that a smooth blended beta floor plus widened trailing-edge sleeve preserves packet safety at higher grid resolution.
15. Packet-local radial support accounting, identifying a mild filled core stretch, narrow annular catch ring, and weak annular skirt as the representative radial support profile.
16. Algebra ledger, separating live angular/current peaks from non-live catch/rematch radial-null peaks.
17. Compact handoff accounting, identifying endpoint-smooth entry/catch/edge support and the entry service gate as the pressure-aware handoff profile for V5 source placement.
18. Full-grid source decomposition, dense source-sector closure, and constrained source-role assignment for radial support, angular capacity, live handoff trim, and current relaxation.
19. Source-family target extraction, including the constant-flux radial string-cloud backbone, endpoint/junction layer, small residual leakage, angular-capacity jacket, and live handoff trim.
20. Beta-memory endpoint receiver accounting, including release-transfer memory, negative-$l$ receiver placement, endpoint transfer accounting, and service-factor retiming anchors.
21. Endpoint source-medium accounting, including the frozen endpoint-J source package, support-edge closure component, source-class classification, current-regulator budget, regulated-medium admissibility, covariant endpoint/source tensor assembly, and constrained field-closure validation.
22. Support stroke/stress accounting, including divergence-form support exchange current, total endpoint/support conservation, support-tail localization, live-exclusion, angular-exchange, and compactness-bracket diagnostics.
23. Reduced endpoint/support principal-symbol diagnostics, including heat/enthalpy response, angular response, support stroke/stress response, director advection, characteristic-speed checks, eigenbasis

completeness, and cone-margin/transport-margin boundary rows.

24. Bounded rapidity transport diagnostics, including raw heat-current stress comparison, rapidity relaxation, row-local rapidity budgets, reduced $1+1$ rapidity advection non-concentration, full-domain fixed-background evolution, service-coordinate aligned-envelope timing, and a local budget limiter guard.
25. Formal source-family equation diagnostics, including regulated endpoint tensor structure, radial director constraints, bounded rapidity law, phase-local source response, service-coordinate release, support-reservoir derivative operator, two-channel support exchange, total-conservation restoration, and fixed-background principal-symbol validation.
26. Cross-surface source-family robustness diagnostics, including baseline/dense V5 surface validation, hard formal-symbol gates, live-support exclusion, total-conservation gates, local exchange-shape constants, support-closure constants, and engineered-surface drift accounting.
27. Fixed-background source-family energy diagnostics, including positive identity symmetrizer, symmetric principal block, in-cone flux margin, lower-order work constants, support-work decomposition, local exchange-shape contribution, support-closure contribution, and available-surface constant stability.
28. Affine-reparameterized radial-null SNEC and finite-window null-accumulation diagnostics comparing live, handoff, endpoint, and source-sector windows with selected radial-null benchmark floors and residual monitors.
29. Causal carrier diagnostics, including shell/throat overlap maps, branch-margin maps, entry reachability, long-tail radial escape, scheduled-ADM probe evolution, dense finite-bundle branch-band transport, exterior-null service-time rating, and chronology-governed service sequencing.

### 17.2 Source-Family Boundary

The prescribed-metric and effective-model evidence below defines the archived V5 design reference. Its endpoint tensor identity, complementary exchange-current fit, and reduced transport estimates retain their respective scopes. Physical source acceptance additionally requires the independent component tensors, corrected causal-frame checks, reciprocal interactions, and joint residual controls in Section 15. The following criteria organize the metric and effective-model reference:

1. live packet norm is safely negative under the selected service factor;
2. packet $\Delta\rho$ and packet-comoving negative-density increments remain quiet;
3. incremental ADM momentum is localized in catch/rematch support infrastructure;
4. 4D demanded-source channels show measurable relief or controlled redistribution, especially in radial-null, angular-pressure, and current channels;
5. point peaks and angular-pressure leakage remain bounded under grid refinement;
6. shell-throat concentration is stable under matched-strength timing and radial-shape perturbations;
7. release, carve, lapse, packet beta-rematch, and packet-local radial-support derivative diagnostics remain bounded under grid refinement;
8. the source-target decomposition assigns the dominant non-live radial body to an explicit radial-tension realization target;
9. the endpoint/junction layer is represented by a finite-width source package with bounded reset-cap support, finite support-edge closure, and live exclusion;
10. the regulated endpoint medium has a small current-regulator budget, positive transport margin, constrained angular response, covariant tensor assembly, and live/collar/packet/current/component exclusion;
11. the support basis fits the complementary endpoint exchange current within its stated residual and localization gates;
12. the regulated director/support-reservoir equation package states the endpoint tensor, radial director, support exchange, bounded rapidity response, phase-local source law, service-coordinate release,

positive split-step transport, and fixed-background principal-symbol gate;
13. the derivative support-reservoir operator preserves the source-family hard gates on the baseline and dense fixed-background ledgers;
14. the reduced endpoint/support principal symbol has real finite in-cone characteristics, complete eigenbases, and positive cone and transport margins on the promoted baseline and dense audit rows;
15. bounded transport rapidity preserves the hard support-sector characteristic gates at observed heat-current stress amplitude;
16. row-local rapidity budgets, reduced advection checks, full-package source-coupling checks, and full-domain fixed-background evolution show that observed support-source rapidity profiles remain below budget after scoped support-edge source-profile normalization, with the limiter guard inactive at observed amplitude;
17. service-coordinate source-release timing supplies an aligned-envelope bound for common nonnegative temporal kernels over service-ordered source bins;
18. cross-surface source-family robustness preserves hard formal-symbol, live-exclusion, and total-conservation gates on the available sealed baseline V5 and sealed dense V5 surfaces;
19. fixed-background energy-estimate certification supplies a positive symmetrizer, symmetric principal block, positive in-cone flux margin, and explicit lower-order support/source work constants on the available complete surfaces;
20. energy-constant auditing classifies the lower-order work constant as a stable limiting theorem constant with measured headroom and a carried safety-margin monitor;
21. reduced coupled-consistency diagnostics evaluate constraint-driver and angular/time response proxies, with physical Einstein–matter evolution subject to the independent tensor and field-equation requirements;
22. large-amplitude support-source stress diagnostics are recorded as engineering-margin classifications, with the full-domain large-stress evolution staying below budget in the sealed beta075 fixed-background model;
23. affine-reparameterized radial-null source-sector windows retain positive benchmark margin with non-affinity and geodesic residual monitoring;
24. dense branch-band finite-bundle diagnostics preserve ray ordering, recovery, radial escape, and bundle-width floor through the carrier-focus collar;
25. exterior-null service-time rating gives a positive operational A-to-B advantage under the selected prescribed schedule;
26. chronology-governed service sequencing maps accepted service edges, return edges, reset intervals, and network cross-links into monotone rail-time advance with safety margin;
27. the component can be stated as a continuous metric expression with clear source-role and causal-carrier assignments.

### 17.3 Acceptance Standard

A prescribed metric and its effective endpoint/operator model are accepted within their stated diagnostic scope when they satisfy the following criteria. Physical load-bearing acceptance additionally closes the common tensor, exchange, field, and coupled-response requirements of Section 15:

1. live packet norm is safely negative under the selected engineered service factor, with out-of-scope service-rating ladders recorded separately as calibration and boundary evidence;
2. packet $\Delta\rho$ and packet-comoving negative-density increments remain quiet;
3. incremental ADM momentum is localized in catch/rematch support infrastructure;
4. 4D demanded-source channels show measurable relief or controlled redistribution, especially in radial-null and current channels;
5. point peaks and angular-pressure leakage remain bounded under grid refinement;
6. shell-throat concentration is stable under matched-strength timing and radial-shape perturbations;

7. release, carve, lapse, packet beta-rematch, compact handoff, and packet-local radial-support derivative diagnostics remain bounded under grid refinement;
8. the source-family target grammar covers the major non-live radial support, endpoint/junction, angular-capacity, current-relaxation, and live-handoff duties, with full-domain realization residuals recorded;
9. the beta075 endpoint source is represented by a frozen finite endpoint package and a regulated anisotropic heat/current medium passing constrained finite-difference field-closure and covariant tensor diagnostics;
10. the support basis supplies a localized complementary exchange-current fit on the baseline and dense audit surfaces;
11. the formal regulated director/support-reservoir source-family equations preserve endpoint tensor closure, support exchange, conservation restoration, angular-exchange absence, support-tail localization, and live support exclusion;
12. the reduced endpoint/support principal-symbol diagnostic has zero hard gate-crossing rows, real finite in-cone characteristic speeds, complete eigenbases, and positive transport margins on the promoted audit rows;
13. bounded rapidity transport clears the observed-amplitude heat/current stress that crosses a raw heat-ratio local gate;
14. rapidity row-budget, reduced advection, full-package source-coupling, and full-domain fixed-background diagnostics preserve observed support-source profiles below budget, with limiter activation reserved for larger over-budget stresses;
15. service-coordinate aligned source timing preserves rapidity budgets for aligned finite pulses and for common nonnegative temporal kernels over service-ordered bins;
16. cross-surface source-family validation preserves the same hard source-family gates on the available baseline and dense V5 surfaces;
17. fixed-background energy-estimate certification supplies positive energy norm, symmetric principal matrix, in-cone flux margin, and explicit lower-order support/source work constants;
18. the lower-order energy constant remains below the configured hard theorem bound and stable across the available dense surfaces, with decomposition recorded by support work, local exchange shape, and support closure;
19. affine-reparameterized radial-null SNEC diagnostics retain positive benchmark margin under the selected source-sector package;
20. causal-carrier diagnostics show service-stage packet protection and live-service radial escape under the selected prescribed-metric probes;
21. dense finite-bundle carrier diagnostics show branch-band recovery with zero caustic-like bundle-collapse flags under the selected prescribed-metric probes;
22. exterior-null service-time ledgers show operational A-to-B advantage under schedule-factor and packet-coordinate transport proxies;
23. chronology-governed service sequencing assigns operational superluminality to a rail-time monotonicity governor and assigns network-scale synchronization to the subsequent safety-engineering layer;
24. reset-domain wake-tail monitors remain exterior to live service and are tracked separately from carrier-band branch behavior;
25. the component can be stated as a continuous metric expression with clear source-role assignments.

## 18. ENGINEERING ACCEPTANCE CRITERIA

The table separates reference-model diagnostics from physical construction requirements. Archived operator margins characterize their specified effective equations; rest-frame-dependent material gates use the corrected classifier. The physical assembly supplies all component tensors and their exchanges, including the reservoir and each replacement residual.

| Criterion | Meaning | Gate |
|---|---|---|
| Packet norm safety | All live packet norm samples remain negative | Required |
| Substrate-separated density | Packet $\Delta\rho$ is small after subtraction | Required |
| Handoff localization | Service $\Delta j_l$ concentrates in catch/rematch and support-shell handoff | Required |
| Reduced support target | Selected support-shell target preserves catch/support fraction, rich packet quietness, and partition closure | Required |
| Continuous metric expression | Support-shell component is represented directly in $\alpha, \beta, G, Q$ as a continuous metric component | Required |
| Matched shell strength | Shape comparisons are made at equal peak $\Delta\beta_{\mathrm{sh,max}}$ | Required for shape selection |
| Radial bearing shape | Raised-cosine annular support shell is preferred under matched-strength comparisons | Required for support-shell source comparator |
| Release choreography | Matched-hold, smooth beta fade, coordinated lapse/carve support, and packet-edge beta rematch preserve packet safety | Required for representative V5 embodiment |
| Packet beta-rematch sleeve | Smooth blended floor and widened trailing-edge sleeve survive grid refinement with zero high-burden live top points | Required for representative V5 embodiment |
| Compact handoff layer | Endpoint-smooth entry/catch/edge windows separate pressure containment from derivative smoothing | Required for representative V5 embodiment |
| Affine radial-null margin | Affine-reparameterized finite-window radial-null exposure is above the selected benchmark floor on live, handoff, endpoint, and source-sector scopes, with non-affinity and radial residual monitors recorded | Required for source-family accounting |
| 4D demanded-source behavior | Radial-null, radial-current, pressure, and packet-comoving density channels remain bounded and improve or redistribute under coupled timing | Required for load-bearing acceptance |
| Source-family target grammar | The role map covers radial backbone, endpoint, angular, live-handoff, and current duties; a physical partition records independent tensors and residuals | Required for source construction |
| Endpoint/junction closure | Endpoint shoulder and reset/decompression cap are represented by a bounded reset-cap body plus finite support-edge closure component with transfer, conservation, and dense-stability diagnostics | Required for endpoint-source acceptance |

| | | |
|---|---|---|
| Endpoint current regulator | A small non-live $\rho + p_l$ regulator removes flux-dominant radial-block rows while preserving current, angular, collar, packet, and closure-component exclusion | Required for beta075 endpoint medium |
| Regulated endpoint medium closure | The fitted tensor carries radial heat/current and angular response with its counted regulator; corrected causal-frame checks determine physical admissibility | Required for endpoint construction |
| Covariant endpoint/source tensor closure | Endpoint components reconstruct the fitted tensor; the geometric J replacement difference is retained separately | Required for tensor accounting |
| Total endpoint/support exchange closure | The complementary exchange-current fit meets active, local, live, tail, and angular residual gates; a physical reservoir also supplies its independent tensor | Required for exchange accounting |
| Complete physical source | Independently counted material and quantum tensors close full-domain Einstein, field, and exchange residuals with both exterior completions specified | Required for physical realization |
| Regulated director/support-reservoir source family | The beta075 source law is represented as a regulated anisotropic heat/current medium entrained to a radial director and coupled to a localized support reservoir with explicit power and radial-force exchange | Required for formal fixed-background source-family acceptance |
| Formal source-family equation package | The fixed-background equation skeleton preserves endpoint tensor identity, director constraints, bounded rapidity response, phase-local source response, service-coordinate release, support exchange, conservation restoration, and inherited principal-symbol gates | Required for source-family validation |
| Cross-surface source-family robustness | The same source-family equations preserve hard formal-symbol, live-exclusion, and total-conservation gates on sealed baseline V5 and sealed dense V5 surfaces | Required for engineered V5 surface robustness acceptance |
| Fixed-background energy-estimate certificate | The source-family equations carry a positive identity symmetrizer, symmetric principal block, in-cone flux margin, and explicit lower-order support/source work constants on the available complete surfaces | Required for fixed-background theorem packaging |
| Lower-order energy constant audit | The support work, local exchange-shape, and support-closure contributions are decomposed, bounded below the hard theorem constant, and stable across the available dense surfaces | Required as a coupled-consistency margin monitor |
| Support compactness bracket | Smaller support-stress bases are retained as brackets only when aggregate closure and localization pass but dense local active closure is tighter than the compact basis | Required for reference selection |

| | | |
|---|---|---|
| Reduced principal-symbol gate | The fixed-background endpoint/support constitutive sector has real finite in-cone characteristic speeds, complete eigenbases, and positive cone and transport margins on promoted audit rows | Required for reduced support-sector transport acceptance |
| Heat-current sensitivity margin | Raw heat-current worsening is stress-tested to identify the first local reset/support-edge boundary before broader PDE claims are made | Required as a margin diagnostic |
| Bounded rapidity transport variable | The heat ratio is parameterized by effective rapidity on its resolved domain; physical boost and diagonal degeneracy use the causal eigensystem | Required for reduced transport modeling |
| Rapidity budget and advection non-concentration | Row-local rapidity budgets and reduced $1+1$ advection preserve observed support-source impulses and source-driven profiles below budget | Required for observed-amplitude support-source acceptance |
| Support-edge source-profile normalization | Scoped entry/catch support-edge source normalization keeps the observed V5 full-package source-coupled evolution below local rapidity budgets | Required for observed-amplitude support-source acceptance |
| Full-domain fixed-background evolution | The sealed endpoint/support package evolves observed support-source profiles across the active non-live support domain with zero live or packet-live evolved rows, zero observed over-budget rows, and inactive observed limiter behavior | Required for reduced full-domain support-source acceptance |
| Service-coordinate aligned source timing | Service-ordered release keeps aligned finite pulses and common nonnegative temporal kernels within row-local rapidity budgets under the split-step fixed-background model | Required for action-level source-timing acceptance |
| Large-source stress margin | Deliberate large-amplitude support-source checks classify engineering headroom; the sealed full-domain large-stress evolution remains below budget in the current fixed-background model | Monitor for future high-amplitude proof obligations |
| Symmetric-hyperbolic source-family packaging | Bounded rapidity, positive transport margins, service-coordinate scheduling, support-reservoir derivative structure, and local margin constants are organized into the accepted fixed-background energy-estimate statement for the tested source family | Required for fixed-background PDE packaging |
| First-order coupled consistency | Reduced constraint-driver and scheduled angular/time response proxies are checked; a physical coupled evolution supplies signed stresses and Einstein–matter equations | Required as a scoped response diagnostic |
| Support-source limiter guard | A local budget limiter remains inactive for observed source amplitude and aligned-envelope checks and activates only for larger over-budget rapidity concentration | Required as a guard for future coupled reduced PDE rungs |

| | | |
|---|---|---|
| Beta-memory receiver | Release-transfer memory is routed into a localized non-live endpoint receiver with packet exclusion preserved | Required for representative receiver grammar |
| Causal carrier reachability | Entry reachability, radial escape, and scheduled probe diagnostics preserve the live service corridor | Required for horizon/carrier acceptance |
| Measured cone classification | Local GZ-like branch behavior is measured as a carrier-band feature and classified by reachability, escape, and scheduled-probe gates | Required for carrier acceptance |
| Dense finite-bundle transport | Branch-band null bundles preserve radial escape, recovery from both-shrinking intervals, ray ordering, and bundle-width floor under the beta-collar rematch setting | Required for carrier acceptance |
| Service-time advantage | Reconstructed restored packet arrival beats the exterior-null reference under schedule-factor and packet-coordinate proxies | Desired for operational rating |
| Rail-time chronology governor | Service edges, return links, cross-links, reset intervals, and ordinary causal edges advance a common rail time with margin | Required for service-network safety engineering |
| Wake-tail restoration | Reset-decompression packet-geometry tails remain exterior to live service and are monitored by forward-domain closure tests | Required for reset-domain acceptance |
| Residual localization and tail audit | Residual maps show localized packet-edge and support-region structure, with outside-support, live, and support-tail fractions recorded for positive support fits | Required |
| $V = 5$ reference robustness | Prescribed service, exchange-fit, effective-operator evolution, and reduced constraint-proxy results retain their tested grids, backgrounds, and margin constants | Required for reference-model acceptance |
| Service-aware extension | Out-of-scope $V = 2.5$ and $V = 10$ ladders identify the need for explicit $V$-aware medium/support closure and high-service causal-margin engineering before a service-family claim is made | Desired future engineering layer |
| Compact absorber law | Catch-edge absorber is expressible as a bounded control law | Required |
| Constraint compatibility | Coarse 3+1 scaffold maintains stable constraint behavior | Required |

## 19. RISK CONTROLS AND DIAGNOSTIC CONTROLS

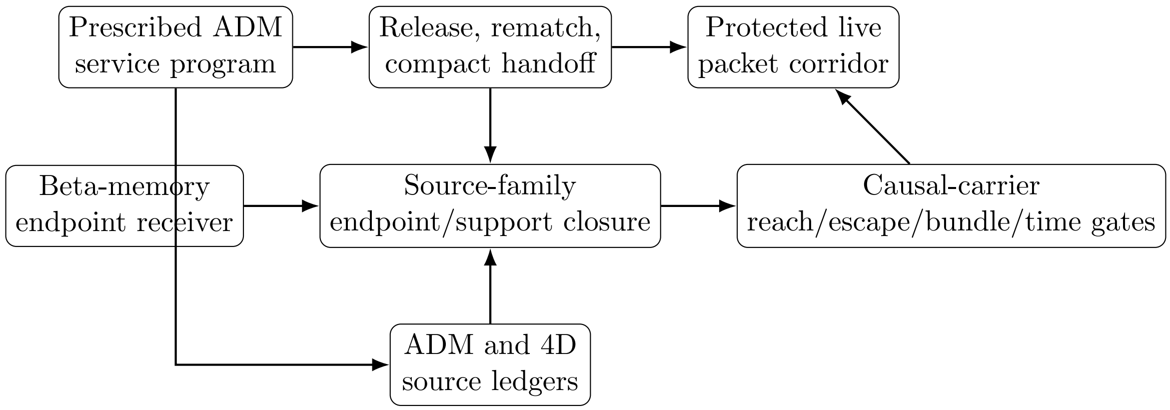
1. **Packet-region density increment.** Packet-region $\Delta\rho$ concentration is controlled by packet-corridor shaping.
2. **Refinement-sharpening catch layer.** Catch-edge sharpening under grid refinement is controlled by absorber broadening and source-family smoothing.

3. **Angular-pressure leakage.** Growth in $p_\Omega$ or angular support entering the packet region is controlled through the throat-capacity/angular jacket sector.
4. **Momentum delocalization.** $\Delta j_l$ spreading across the packet interior is controlled by the handoff model and entry service gate.
5. **Single-channel additive burden.** Carrying-flow amplitude that adds radial-null or radial-current burden is paired with coupled clock-lapse, rail-stretch, or throat-capacity refinement.
6. **Shell-throat overlap concentration.** Added radial-null, radial-current, or angular-pressure burden where $W_{\mathrm{sh}}$ overlaps $G, Q$ gradients is controlled by timing broadening and intrinsic-metric partner refinement.
7. **Point-peak growth.** Local peak growth is controlled by timing broadening, strength selection, and source-family smoothing.
8. **Radial bearing over-broadening.** Support-shell annulus spread into packet-sensitive structure is controlled by the localized raised-cosine bearing band and narrower comparator families.
9. **Packet beta-rematch seam dependence.** Floor/sleeve transition sensitivity is controlled by smooth blended rematch shaping.
10. **Smooth-rematch margin loss.** Narrow trailing-edge support is controlled by radial rematch broadening and pressure-aware handoff smoothing.
11. **Service-factor boundary behavior.** Out-of-scope service-rating ladders are treated as diagnostics for V-aware engineering. The $V = 2.5$ ladder is source-safe but exposes service-independent medium/support closure calibration debt. The $V = 10$ ladder fails live packet source safety and identifies the high-service causal-margin boundary of the present package.
12. **Radial scaffold realization.** The dominant non-live radial infrastructure burden is controlled by expressing it as a constant-flux radial-tension backbone with explicit residual trims.
13. **Endpoint/junction concentration.** Support-edge shoulder and reset/decompression cap burdens are controlled by a frozen finite endpoint source package with bounded reset-cap support, finite support-edge closure, transfer accounting, concentration checks, covariant tensor closure, and dense-stability diagnostics.
14. **Endpoint current and angular response.** Flux-dominant radial-block rows are controlled by a small non-live current regulator, while the non-live angular-pressure channel is controlled as an internal response of the same regulated anisotropic heat/current medium.
15. **Support exchange completion.** Endpoint exchange current is controlled by a phase-aware support stroke/stress-potential sector whose divergence supplies the complementary support current, with active, allowed, local, live, tail, and angular-exchange residuals audited on baseline and dense ledgers.
16. **Formal source-family structure.** The effective endpoint/support package specifies endpoint tensor structure, radial director constraints, bounded heat-ratio response, phase-local forcing, scheduled release, and two-channel exchange. Physical completion supplies an independent reservoir tensor and verifies its reciprocal exchange.
17. **Support-reservoir exchange shape.** Local support-reservoir power and radial-force channels are controlled by derivative support-reservoir validation and carried as explicit local exchange-shape constants in the fixed-background source-family model.
18. **Cross-surface source-family robustness.** Available-surface robustness is controlled by rerunning the same formal source-family validation on sealed baseline V5 and sealed dense V5 surfaces, with hard formal-symbol, live-exclusion, total-conservation, and drift gates recorded. Non-V5 service-rating ladders are kept as design maps for V-aware source-law extension.
19. **Heat-current cone-margin fragility.** Boundary-near reset/support-edge rows are controlled by reduced principal-symbol sensitivity checks, regulator-margin monitoring, and bounded rapidity transport.
20. **Rapidity source concentration.** Support-source rapidity impulses and source-driven profiles

are controlled by row-local budgets, reduced advection non-concentration checks, scoped support-edge source-profile normalization, full-domain fixed-background evolution, service-coordinate aligned source timing, and a limiter guard that remains inactive for observed-amplitude source kicks.

21. **Large-source margin classification.** Deliberate large-amplitude support-source checks are recorded as engineering-margin classifiers. Package-level phase-boundary slices identify tight local margins, while the sealed full-domain fixed-background large-stress evolution remains below budget in the current model.
22. **Source-time envelope.** Support-source release is controlled by service-coordinate ordering and common nonnegative temporal kernels over service-ordered bins. Bounded service-local timing jitter is the tolerance class for phase-error characterization around that aligned envelope.
23. **Energy-estimate packaging.** Symmetric-hyperbolic packaging is controlled by fixed-background bounded rapidity assumptions, positive transport margins, support-reservoir derivative structure, service-coordinate source response, positive identity symmetrizer, symmetric principal block, in-cone flux margin, and recorded lower-order support/source work constants.
24. **Lower-order energy constant.** The fixed-background work constant is controlled by explicit decomposition into support work, local exchange shape, and support closure, with dense-surface recurrence and hard-bound headroom recorded as the safety-margin monitor for first-order coupled consistency.
25. **First-order coupled consistency.** Reduced constraint-driver and response proxies retain their effective-model scope. The physical coupling gate evaluates independently supplied stress, exchange, field equations, and metric constraints.
26. **Receiver timing.** Beta-memory endpoint receiver action is controlled by release-transfer memory, negative-$l$ localization, packet exclusion, and service-factor retiming anchors.
27. **Causal carrier branch structure.** Local radial branch behavior is controlled by reachability, radial escape, shell/throat overlap, scheduled-ADM probe diagnostics, and dense finite-bundle carrier audits.
28. **Carrier-focus collar compression.** Finite-bundle compression in the minus-branch carrier band is controlled by widening and timing the trailing-edge beta-rematch collar until dense bundle transport remains above the caustic-like collapse thresholds.
29. **Operational service-time rating.** A-to-B service timing is controlled by an exterior-null comparison ledger that records prepared-service and request-triggered service clocks, schedule-factor distance, packet-coordinate distance, and the strict $l = \sigma$ centerline control.
30. **Chronology-governed service sequencing.** Service arming is controlled by a rail-time governor that advances entry, carrier, return, cross-link, reset, endpoint-readiness, and multi-leg routing intervals with configured margin.
31. **Wake-tail restoration.** Reset-domain packet-geometry tails are controlled by explicit post-service wake-tail monitors and domain-extension closure tests.
32. **Constraint stability.** Coarse 3+1 scaffold constraint residuals are controlled through smoothing, reparameterization, and source-ledger redesign.

## 20. ILLUSTRATIVE SYSTEM DIAGRAM




## 21. CLAIM-ORIENTED EMBODIMENTS

The following claim-oriented embodiments clarify the technical scope of the disclosure.

1. A method of operating a prescribed spacetime metric service, comprising preparing a standing support substrate, defining a live packet corridor within a larger support shell, routing a carrying-flow field through the support shell, and applying a locked catch/rematch control window to localize service-induced momentum demand near a packet edge.
2. The method of item 1, wherein the live packet corridor has radius $R_{\text{pass}}$ smaller than a support radius $R_{\text{th}}$, and wherein a radial support-edge width $w_{\text{th}}$ is selected to reduce radial pressure burden while a lapse cushion preserves packet causal margin.
3. The method of item 1, wherein a dimensionless service factor $V$ sets the scheduled carrying-flow level and packet-frame comparison schedule, and wherein $V$ is used to define an engineered operating point and to organize out-of-scope service-rating diagnostic ladders.
4. The method of item 3, wherein an engineered operating embodiment is rated at $V = 5$, and wherein service factors outside that engineered scope are evaluated by fresh service-rating diagnostic ladders before promotion to an operating embodiment.
5. The method of item 1, wherein an ADM diagnostic ledger is computed from fields $\alpha$, $\beta$, $G = \gamma_{ll}$, and $Q = \gamma_{\Omega\Omega}$, and wherein the ledger includes $\rho_{\text{ADM}}$ and $j_l$.
6. The method of item 5, further comprising subtracting a time-matched standing substrate from a live service field to identify $\Delta\rho$ and $\Delta j_l$.
7. The method of item 6, wherein a service pass is accepted when $\Delta\rho$ is small inside the live packet and $\Delta j_l$ is localized near a catch/rematch handoff layer.
8. A spacetime metric engineering system comprising a standing support shell, an angular jacket, a clock-lapse cushion, a support-contained carrying-flow field, a live packet corridor, and a packet-edge catch/rematch absorber.
9. The system of item 8, wherein the packet-edge catch/rematch absorber comprises a kinematic handoff law multiplied by a compact inner-edge spatial envelope centered near $\xi = -R_{\text{pass}}$.
10. The system of item 9, wherein a residual-responsive limiter bounds the absorber amplitude.
11. The system of item 8, wherein a support-shell overlay is applied through an annular timing window $W_{\text{sh}}$ and modifies one or more of $\beta$, $\alpha$, $G$, and $Q$.
12. The system of item 11, wherein the support-shell overlay includes

$$\beta = \beta_0 + a_\beta W_{\text{sh}}, \quad \alpha = \alpha_0 e^{\kappa_\alpha W_{\text{sh}}}, \quad G = G_0 e^{\kappa_G W_{\text{sh}}}, \quad Q = Q_0 e^{\kappa_Q W_{\text{sh}}}.$$

13. The system of item 12, wherein $W_{\text{sh}}$ includes a raised-cosine annular radial bearing function centered in the support-shell infrastructure band.

14. The system of item 13, wherein support-shell shape comparisons are normalized by a matched peak carrying-flow perturbation $\Delta\beta_{\mathrm{sh,max}}$.
15. The method of item 1, further comprising a release-choreography law that holds the packet in a matched state, fades beta with a finite smooth profile, maintains lapse/carve support during the fade, and relaxes support after packet-edge mismatch is reduced.
16. The method of item 15, wherein a packet-local beta rematch correction is applied through a trailing-edge sleeve and a smooth blended center floor.
17. The method of item 16, wherein the trailing-edge sleeve is radially widened until the blended floor survives grid refinement with all live packet-norm samples safely negative and zero top-channel live packet points at the engineered service factor.
18. The method of item 15, further comprising a compact handoff layer with endpoint-smooth entry containment, catch/rematch bearing, and trailing-edge smoothing windows.
19. The method of item 18, wherein the compact handoff layer assigns live radial-pressure containment to the entry window and angular/current plus radial-null derivative relief to catch and edge windows.
20. The system of item 8, further comprising a packet-local radial support profile with a mild filled core stretch, a narrow annular catch ring, and a weak annular skirt in $\gamma_{ll}$.
21. The system of item 8, further comprising a source-family target grammar that assigns demanded-source burden to a constant-flux radial string-cloud backbone, a coupled endpoint/junction layer, small core-body residual leakage, live handoff trim, angular-capacity support, and non-live current relaxation.
22. The system of item 21, wherein the constant-flux radial string-cloud target has positive energy density $\rho = \Phi/\gamma_{\Omega\Omega}$, radial tension close to $p_l = -\rho$, small transverse pressure and current, and nearly constant areal flux $\Phi$.
23. The system of item 21, wherein the endpoint/junction layer carries support-edge shoulder trim and reset/decompression cap demand in one coupled finite-width source target.
24. The system of item 23, further comprising a beta-memory endpoint receiver whose amplitude is tied to accumulated beta-release transfer and whose active support is localized in a non-live negative-$l$ support-edge receiver band.
25. The system of item 24, wherein a beta100 release-timing stress receiver setting uses the same receiver grammar with service-specific post-release retiming to preserve selected, current, and angular endpoint transfer accounting.
26. The method of item 1, further comprising affine-reparameterized radial-null accounting using a coordinate-time radial null tangent, measured non-affinity, affine window weighting, and radial geodesic residual monitors.
27. The method of item 1, further comprising causal-carrier accounting using radial null speeds $v_\pm = -\beta \pm \alpha/\sqrt{\gamma_{ll}}$, branch-margin maps, measured cone classification, entry reachability, radial escape, and scheduled probe evolution.
28. The method of item 27, wherein reset-domain packet-geometry wake-tail traces are monitored separately from live-service carrier acceptance and remain exterior to the live packet corridor during the scheduled probe audit.
29. The method of item 27, further comprising a dense finite-bundle carrier audit in which recovered branch-band seeds are expanded into radial null bundles, and wherein a trailing-edge beta-rematch collar is widened or timed until the bundles preserve radial escape, recovery from both-shrinking intervals, ray ordering, and a positive bundle-width floor.
30. The method of item 29, wherein the dense finite-bundle carrier audit is treated as a CHL-adjacent branch-band optics gate for the throat-loaded active-rail service architecture.
31. The method of item 27, further comprising an exterior-null service-time advantage ledger in which exterior endpoints A and B are fixed, packet departure and restored arrival events are recorded, prepared-service and request-triggered clocks are evaluated, and schedule-factor and

packet-coordinate transport proxies are compared with an exterior null reference time.

32. The method of item 31, further comprising a rail-time chronology governor in which each armed service edge, return edge, cross-link, reset interval, and ordinary causal edge advances a common rail time with safety margin, and in which connected rail networks are scheduled by one chronology policy.
33. The system of item 23, wherein the endpoint/junction layer is represented by a frozen finite endpoint source package comprising a bounded reset-decompression cap body and a finite support-edge closure component, each excluded from the live packet corridor.
34. The system of item 23, further comprising a non-live endpoint current regulator applied in the radial $\rho + p_l$ pair, wherein the regulator makes a radial heat/current block boost-diagonalizable while preserving independent-current, angular, collar, packet, and conservation-closure-component exclusion.
35. The system of item 23, wherein the endpoint/junction layer is carried by a regulated anisotropic heat/current medium with an internal angular-pressure response and a positive transport-margin diagnostic under finite-difference field-closure validation.
36. The system of item 23, further comprising a covariant identity for the fitted endpoint tensor and a separately recorded difference from the geometric endpoint target.
37. The system of item 23, further comprising a phase-aware support basis fitting a complementary endpoint exchange current, with an independent reservoir tensor as a physical realization requirement.
38. The system of item 37, wherein total endpoint/support exchange closure is accepted when active, allowed, local, live-leakage, outside-support, support-tail, and angular-exchange residuals remain within the selected baseline and dense audit gates.
39. The system of item 37, wherein a compact support-stress basis is retained as a compactness bracket when aggregate conservation and localization pass while dense local active closure selects a larger support-stress reference basis.
40. The system of item 37, further comprising a reduced fixed-background endpoint/support principal-symbol diagnostic that perturbs heat/enthalpy, angular response, support stroke/stress response, and director advection variables and accepts rows with real finite in-cone characteristic speeds and complete eigenbases.
41. The system of item 40, wherein the regulated heat/current transport response is represented by a bounded rapidity variable whose perturbations preserve subluminal heat-flux ratio and positive cone-margin gates at observed stress amplitude.
42. The system of item 41, further comprising row-local rapidity budgets, a reduced $1 + 1$ rapidity advection non-concentration diagnostic, full-domain fixed-background source evolution, and a local budget limiter guard reserved for larger over-budget support-source concentration.
43. The system of item 41, further comprising a service-coordinate source-release law and an aligned-envelope rapidity certificate that bounds common nonnegative temporal kernels over service-ordered source bins.
44. The system of item 41, further comprising a regulated director/support-reservoir source-family equation package in which a regulated anisotropic heat/current endpoint medium is entrained to a radial director and coupled to a localized support reservoir.
45. The system of item 44, wherein a physical reservoir specialization supplies an independent tensor with the required power and radial-force exchange and counted stored energy, momentum, and strain.
46. The system of item 44, further comprising cross-surface source-family robustness validation on sealed baseline V5 and sealed dense V5 source-family surfaces.
47. The system of item 44, further comprising symmetric-hyperbolic or energy-estimate packaging of the fixed-background source-family equations using bounded rapidity, positive transport margins, service-coordinate source response, support-reservoir derivative structure, and recorded local margin constants.

48. The system of item 44, further comprising a fixed-background energy-estimate certificate with positive identity symmetrizer, symmetric principal block, in-cone flux bound, and explicit lower-order work constants for support exchange and scheduled source response.
49. The system of item 47, further comprising an energy-constant audit that decomposes the lower-order work constant into support-work, local exchange-shape, and support-closure contributions and carries the resulting theorem constant into first-order coupled consistency diagnostics.
50. A numerical source-accounting process comprising constructing an exact field builder for $\alpha$, $\beta$, $G$, and $Q$; computing ADM constraint fields; performing standing substrate subtraction; computing a 4D demanded-source ledger; recording release/carve/lapse/beta-rematch/compact-handoff derivative diagnostics; computing semilocal finite-window null-accumulation margins; extracting source-family targets; evaluating endpoint/junction receiver transfer; classifying endpoint source blocks; applying a non-live current regulator; validating a regulated endpoint heat/current medium; assembling a covariant endpoint/source tensor; fitting a divergence-form support stroke/stress-potential sector; evaluating total endpoint/support closure against component-specific source roles; computing reduced endpoint/support principal-symbol characteristics; evaluating bounded rapidity transport response; auditing row-local rapidity budget and advection non-concentration; evolving full-domain fixed-background support-source profiles; certifying service-coordinate aligned source-release envelopes; validating regulated director/support-reservoir source-family equations; auditing cross-surface source-family robustness; certifying fixed-background source-family energy estimates; and auditing lower-order energy constants.

## 22. DIAGNOSTIC DATA PRODUCTS

Data products for the engineered $V = 5$ operating embodiment include:

1. the component cross-reference and joint-coordination specification, source-role allocation audit, counted support and integrated construction gates, and corrected classifier evidence, with their registered construction backgrounds;
2. exact-builder ADM implementation package;
3. substrate-subtracted $\rho_{\mathrm{ADM}}$ and $j_l$ maps;
4. reduced support-shell ansatz ledger and selected support-shell target ledger;
5. generalized diagnostic ladder with target/ramp separation and service-factor input generation;
6. historical V10 selected-target release-timing stress comparison for reduced small-target channel-placement context;
7. continuous support-shell 4D demanded-source ledger path;
8. coupled timing sweep for carrying-flow, clock-lapse, and rail-stretch support-shell partners;
9. throat-capacity same-window ledger;
10. support-shell shape-selection and matched-strength robustness ledger;
11. release-choreography ledger;
12. release-choreography floor-union refinement check showing preservation of the blended, widened trailing-edge V5 family under higher grid resolution;
13. compact handoff, entry service gate, and channel-cause ledgers for pressure-aware entry/catch/edge shaping;
14. full-grid source-decomposition and dense source-sector closure ledgers;
15. source-family target extraction for the constant-flux radial string-cloud backbone, endpoint/junction layer, residual leakage, angular capacity, live handoff trim, and current relaxation;
16. beta-memory endpoint receiver transfer-accounting and retiming ledgers;
17. frozen endpoint-J source ledgers for the bounded reset-cap body and finite support-edge closure component;
18. endpoint source-class, current-regulator, regulated-medium admissibility, covariant endpoint/source

tensor, and constrained field-closure ledgers;

19. support stroke/stress-potential exchange ledgers, total endpoint/support conservation ledgers, compact support-stress bracket ledgers, and outside/live/support-tail localization audits;
20. support-sector robustness, reduced endpoint/support principal-symbol, and principal-symbol sensitivity ledgers;
21. bounded rapidity transport pilot, row-local rapidity budget, reduced $1+1$ rapidity advection non-concentration ledgers, full-package support-source coupling ledgers, support-edge source-profile normalization ledgers, full-domain fixed-background evolution ledgers, service-coordinate aligned schedule ledgers, and aligned-envelope source-timing certificates;
22. formal source-law definition, formal source-family equation package, source-family validation, discretization robustness, timing-jitter, cross-surface source-family robustness, fixed-background energy-certificate, and energy-constant audit ledgers;
23. causal-carrier reachability, radial escape, shell/throat branch-margin, scheduled-ADM probe, and dense finite-bundle carrier ledgers;
24. tensor-consistent beta-collar generator ledgers for the conservative $rematch\_w6\_t1p5$ carrier lead and the wider $rematch\_w8\_t2p0$ optical-relief bracket;
25. exterior-null service-time advantage ledgers comparing prepared-service and request-triggered service clocks with schedule-factor and packet-coordinate transport proxies;
26. chronology-governance source materials and service-sequencing safety specifications for rail-time monotonicity across single-rail and connected-network operation;
27. affine-reparameterized radial-null SNEC and semilocal finite-window null-accumulation diagnostics comparing selected radial-null exposure windows with a selected benchmark floor and residual monitors;
28. service-rating ladder diagnostics at $V = 2.5$ and $V = 10$ showing that $V = 5$ is the current engineered operating point, that the non-V5 ladders are outside engineered scope, and that service-aware medium/support closure plus high-service causal-margin engineering are the next source-law obligations.

## 23. CONCLUSION

The active-rail system organizes spacetime metric engineering as a source-placement and causal-carrier architecture with an explicit source-family target grammar. The engineered $V = 5$ operating embodiment separates standing substrate curvature from live service demand and localizes selected support-shell and handoff increments into catch/rematch, support-edge, and endpoint infrastructure while preserving packet safety. The current service-rating diagnostics at $V = 2.5$ and $V = 10$ are outside the engineered V5 scope and serve as design maps for the next V-aware source-law layer, while beta100 receiver retiming supplies a release-timing stress anchor.

The metric controls remain componentized: carrying-flow $\beta$ expresses service load, clock-lapse $\alpha$ protects timing and causal margin, rail-stretch $G$ controls radial metric capacity, and throat-capacity $Q$ supplies angular geometry. Their coordinated windows localize demands in the recorded service tests. The source grammar retains standing radial support, angular capacity, endpoint and live-handoff response, current relaxation, and stored support state. Their physical realizations interact through shared geometry, material fields, and exchange.

The archived beta075 reference contains a reproducible geometric demand, a fitted anisotropic endpoint tensor, a complementary support exchange-current fit, and fixed-background transport and energy estimates. The sealed designation preserves those parameters and operator calibrations. Physical assembly supplies an independent reservoir tensor, the endpoint replacement difference, all remaining component stresses, and corrected rest-frame-dependent admissibility. The effective operator estimates continue to provide regression controls at their documented scope.

The source tests distinguish the bulk tension of a finite throat from its signed null-opening requirement. In the smooth semiclassical seed the relaxed condensate carries 0.612% of throat tension, while the tested neutral scalar also has an independent, large opening deficit. A coupled construction therefore gives bulk support an explicit counted allocation and evaluates quantum supply in both null directions across the transitions. Angular response, preload, optical confinement, restoring forces, and reset storage participate in the same balance.

The construction sequence specifies candidate fields and their roles, applies necessary tensor and integral screens, solves the standing geometry/material/quantum system, evaluates coupled response, and then admits a bounded service cycle. Source-derived trajectories are compared with the established packet, carrier, and timing constraints. Common tensor accounting, reciprocal forces, and final reservoir readiness connect the original differentiated architecture to its physical source-selection problem.

## 24. SOURCE CODE AND ARTIFACT REPOSITORY

The version-controlled source code, supporting reports, disclosure source, and project history for this work are maintained at https://github.com/dgoldman0/spacetime-metric-engineering. Large generated run-data artifacts are not represented by this repository link.

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